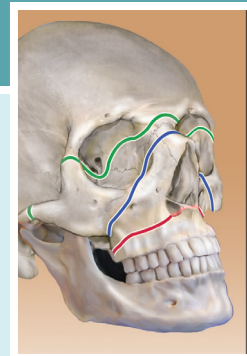


The Female Athlete

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THE GROWING IMPORTANCE AND RECOGNITION OF THE FEMALE ATHLETE

Over the past several decades there has been a rapid development of competitive sporting events for women, and parallel to this, an emergence of increasing numbers of excellent competitive women athletes.

Before the 1970s, few women participated in organized sport. However, the passage of Title IX of the Educational Assistance Act of 1972¹ required institutions receiving federal money to offer equal opportunities to both males and females in all programs including athletics. This sparked a rapid growth not only in collegiate women's sport opportunities, but also in those sport opportunities available to high school and recreational female athletes (Table 27.1).

The female athlete market is now a major target for businesses as exemplified by women's clothing. Women's sporting gear before the 1980s was difficult to find. However, with increased opportunities in sports for women following the passage of Title IX, the demand for female-specific sport clothes and equipment has increased. Prior to this time, female athletes often wore men's shoes (a practice that was associated with an increase in foot problems), men's warm-ups, and sport protection equipment (shin guards in soccer, eye protection in racket sports, mouth guards, etc.). However, in the 1990s the business of women's sport gear—including warm-ups, shorts, skirts, shirts, sports bras, protective pads, and shoes—grew rapidly. Women traditionally spend more money on all clothing needs compared to men, and sport wear has become a fashion statement for women. Sized to the women's figure, braces and other protective sport gear now have a more comfortable fit. Lightweight clothing that wicks away perspiration is prevalent and comes in pinks, purples, and other bright colors for stylish looks. Shoes made for women are now a more comfortable “fit,” which has resulted in fewer calluses, corns, and other foot issues.

Before 1970, rarely were the results of women's sports contests found in newspapers or within the pages of *Sports Illustrated* or on nightly TV news or radio; however, this trend is changing.

Women's golf and tennis events are now aired on primetime television. The Women's National Basketball Association (WNBA) aired 45 regular season games in 2017. Fox reports that 6 of the top 10 searches of 2016 Olympic Athletes were for females.⁵

Parallel and (perhaps one could argue) secondary to the increased emphasis on women's sport participation, there has

been an improvement in women's sport performance in swimming and running events, and the speed of basketball, soccer, tennis, and volleyball games has increased with the improvement in women's skills and overall athleticism.

Not only have the numbers and recognition of highly competitive female athletes increased over the last 50 years, but also the prevalence of women of all ages participating in recreational athletics has skyrocketed, making it essential that those caring for the female athlete appreciate her uniqueness as gender-based differences are important when developing novel approaches to prevention, diagnosis, and treatment of injury and illness.⁶ The term “sexual dimorphism” is defined as: “The condition in which the two sexes of the same species exhibit different characteristics beyond the differences of their sexual organs.”⁷ In humans, these differences are related to the expression of genes on the X and Y chromosomes. These differences form the basis of this chapter and include not only anatomic and physiologic differences (Table 27.2) but also certain aspects of illnesses and injuries unique to the female athlete.

GENERAL CONSIDERATIONS

Conditioning

Conditioning has been defined as “a process in which stimuli are created by an exercise program performed by the athlete to produce a higher level of function.”⁹ A properly constructed conditioning program should maximize performance and minimize the chance of injury. There have been ongoing debates as to whether conditioning techniques used for males are appropriate for females. Little boys start out wishing to do weights with their fathers or older brothers or friends. This is infrequently true in girls. Hence, in the mid to late teenage years, boys are typically familiar with weight workouts and have already incorporated strengthening programs into their sport conditioning programs. At this time, girls most likely are just beginning to learn that a weight strengthening program is a needed part of a well-constructed conditioning program. Some theorize that girls and young women may be leery of participating in a weight training program for fear of developing bulky muscles. They should be reassured that strength training does not necessarily have to involve lifting heavier and heavier weights and is associated with not only better performance but also a decrease in injury.^{9,10}

Static and dynamic balance are vital for maximizing sport performance as well as enhancing injury prevention.¹¹ Articles

Abstract

Over the past several decades there has been a rapid development of competitive sporting events for women, and parallel to this, an emergence of increasing numbers of excellent competitive women athletes. Not only have the numbers and recognition of highly competitive female athletes increased over the last 50 years, but also the prevalence of women of all ages participating in recreational athletics has skyrocketed, making it essential that those caring for the female athlete appreciate her uniqueness as gender-based differences are important when developing novel approaches to prevention, diagnosis, and treatment of injury and illness. The term sexual dimorphism is defined as: “The condition in which the two sexes of the same species exhibit different characteristics beyond the differences of their sexual organs” (Wikipedia 2017 https://en.wikipedia.org/wiki/sexual_dimorphism#External_links). In humans, these differences are related to the expression of genes on the X and Y chromosomes. These differences form the basis of this chapter and include not only anatomic and physiologic differences but also certain aspects of illnesses and injuries unique to the female athlete.

Keywords

female
athlete
foot/ankle
concussion
triad
knee
shoulder

TABLE 27.1 Growth in Women's Sports^{a,b}

Participation Level	NUMBER OF WOMEN PARTICIPATING			
	1971–1972	2000–2001	2010–2011	2015–2016
High school	294,015	2,784,154	3,173,549	3,324,326
Collegiate	29,972	150,916	190,000	216,286
Olympic	1264 (17.5% of Olympians)	4935 (37.5% of Olympians)	5091 (42.4% of Olympians)	5800 (43.5% of Olympians)

www.nfhs.org; www.ncaa.org; www.olympic.org.

TABLE 27.2 Anatomic and Physiologic Gender Differences^{a,b}

Parameter	Postpubertal Girls	Postpubertal Boys	Impact
Oxygen pulse (efficiency of cardiorespiratory system)	Lower	Higher	Higher oxygen pulse provides boys an advantage in aerobic activity.
VO _{2 max} (reflects level of aerobic fitness)	Lower	Higher	Boys have greater aerobic capacity.
Metabolism (basal metabolic rate [BMR])	6%–10% lower (when related to body surface area)	6%–10% higher (when related to body surface area)	Girls need fewer calories to sustain the same activity level as boys.
Thermoregulation	Equals boys	Equals girls	They have equal ability to adequately sweat in a hot environment to decrease core body temperature.
Endocrine System			
Testosterone	Lower	Higher	Boys have increased muscle size, strength, and aggressiveness.
Estrogen	Higher	Lower	Unknown if it is related to increase in ligamentous laxity or in rate of ACL injuries.
Height	64.5 inches	68.5 inches	The increased height and weight in boys give them structural advantages.
Weight	56.8 kg	70.0 kg	
Limb length	Shorter	Longer	Boys can achieve a greater force for hitting and kicking.
Articular surface	Smaller	Larger	May provide boys with greater joint stability; boys have greater surface area to dissipate impact force.
Body shape	Narrower shoulders Wider hips Legs 51.2% of height More fat in lower body	Wider shoulders Narrower hips Legs 52% of height More fat in upper body	Girls have lower center of gravity and therefore greater balance ability; girls have increased valgus angle at the knee that increases knee injuries; boys and girls have different running gaits.
%Muscle/total	–36%	–44.8%	Boys have greater strength and greater body weight ^a speed.
%Fat/total body weight	–22% to 26%	–13% to 16%	Girls are more buoyant and better insulated; they may be able to convert fatty acid to metabolism more rapidly.
Age at skeletal maturation	17–19 years	21–22 years	Girls develop adult body shape/form sooner than boys.
Cardiovascular system			Stroke volume in girls is less, necessitating an increased heart rate for a given submaximal cardiac output;
Heart size	Smaller	Larger	cardiac output in girls is 30% less than in boys; the risk of hypertension may be less in girls.
Heart volume	Smaller	Larger	
Systolic blood pressure	Lower	Higher	
Hemoglobin		10%–15% > per 100 mL blood	The oxygen carrying capacity of blood is greater in boys.
Pulmonary System			
Chest size	Smaller	Larger	Total lung capacity in boys is greater than in girls.
Lung size	Smaller	Larger	
Vital capacity	Smaller	Larger	
Residual volume	Smaller	Larger	

^aThere are no appreciable differences in these parameters prior to puberty; therefore prepubertal boys and girls can compete on a fairly equal basis.

ACL, Anterior cruciate ligament.

From Yurko-Griffin LY, Harris S. Female athlete. In: Sullivan JA, Anderson SJ, editors. *Care of the Young Athlete*. Rosemont, IL: American Academy of Orthopaedic Surgery; 2000:138–148 with permission.

have been written and comments made in scientific meetings regarding the need, or the lack of need, to alter training and conditioning in several women's sports such as soccer, softball, basketball, volleyball, and lacrosse.¹² Suggestions have been made for players to incorporate more agility and balance exercises to try to decrease the frequency of noncontact anterior cruciate ligament (ACL) injuries in these sports.¹³⁻¹⁵ The adage "stronger is better," is now being modified since strong muscles that fire at inappropriate times can cause harm rather than ensure protection from injury during an athletic event. Programs for women's sports where ACL risks are high are adopting training and conditioning routines that emphasize not only the traditional strength, flexibility, and aerobic conditioning exercises, but also they have drills aimed to enhance balance and agility.^{14,16-20} Incorporating plyometric drills appears to be appropriate in both male and female sport programs. (See section on ACL injuries in women in this chapter.)

Greater emphasis on core strengthening may also be appropriate for females who have been reported to have more anterior pelvic tilt than men, a trait that is linked with patellofemoral pain syndrome.¹³ Medial quadriceps exercises are recommended for females with laterally tracking patellae especially those with increased Q angle, increased tibial tubercle to tibial groove distance, and those with increased knee valgus.²¹ These exercises should commence during the prepubescent age and continue through the pubertal years and into post pubertal maturation. By building strength in the vastus medialis, hopefully the patella will sit more centrally in the trochlear groove decreasing patellofemoral pain by balancing patellar trochlear forces.

Since postpubertal females are reported to have increased flexibility compared to males, strengthening scapular stabilizers and the muscles of the rotator cuff to prevent shoulder instability issues is recommended in swimmers; in volleyball, lacrosse, and softball players; and in those playing racket sports.²² Prevention rather than "cure" should be the norm and has been proven to be beneficial. Following puberty, training and conditioning programs should also account for inherent gender-based, physical, and physiologic differences (see Table 27.2). Males are more fully developed in the upper body with a narrow pelvis and hence, a higher center of gravity; whereas females have narrower shoulders compared to males, but proportionally a wider pelvis resulting in a lower center of gravity, but they have a more difficult time achieving the upper body strength of their male counterparts. Males have more muscle mass per body weight; whereas female athletes, even those who are considered to be well conditioned, have greater body fat than males (18% to 20% females vs. 10% to 15% males). Moreover, males have a greater thoracic capacity and hence, a greater VO₂ max than do females. Overall aerobic capacity in females is less than that of males following puberty. Hence, one notes that although female athletes have improved significantly over the last 10 years, their times in anaerobic and aerobic running events are still not equal to those of males (Table 27.3). A few "conditioning tips" for female athletes are listed in Table 27.4.

In the last decade there has been an emphasis on screening athletes at all levels of play from those who exercise for health and fitness to recreational players to highly trained competitive

TABLE 27.3 Improvements in Women's Anaerobic and Aerobic Running Events^{a,b}

Collegiate Track	1972	1998	2012	2017
800 m	2:04:7	2:06:30	2:03:34	2:02:36
3200 m	10:51:0	10:25:99	10:08:11	10:00:13
10,000 m	33:36:51	32:56:63	32:41:63	32:38:57
Boston Marathon	1972	1996	2011	2017
	3:10:26	2:27:12	2:22:36	2:21:52

www.ncaa.org; www.baa.org.

TABLE 27.4 Conditioning Tips for Recreational Athletes^{a,b}

- Emphasize core strength to minimize stress on lower extremities.
- Emphasize strengthening of scapular stabilizers and muscles involved in dynamic stabilization of the glenohumeral joint to minimize laxity issues of the shoulder joint.^a
- Emphasize vastus medialis obliques (VMO) strength when doing lower extremity strengthening exercises to improve patellar tracking.^b
- Minimize loading the patellofemoral joint in a fully flexed knee position, that is, consider short arc extension and leg press exercises in place of squats, lunges, and full arc extension exercises.^b
- Perform upper extremity strengthening exercises at shoulder height and below to minimize stress on the rotator cuff (pull downs, overhead dumbbell press, etc.).

^aKibler WB, Sciascia AD, Uhl TL, Tambay N, Cunningham T. Electromyographic analysis of specific exercises for the scapular control in early phases of shoulder rehabilitation. *Am J Sports Med*. 2008;36:1789.

^bwww.healthline.com/health/fitness-exercises/vastus-medialis-exercises.

athletes for functional movement skills and balance to detect deficits that could predispose the athlete to injury. Exercises to correct these deficits can then be incorporated into the athlete's exercise program not only to improve physical performance parameters (e.g., increased aerobic power, strength, speed, agility, and balance), but also to prevent injury. Such programs have proven to be very highly effective in achieving both of these goals.^{11,26-30}

Nutrition and Hydration

Sports nutrition is currently recognized as one of the most important aspects of improving sports performance. Inadequate nutritional intake has been reported to be more common in female than male athletes. Both the content and timing of optimal nutrition may significantly impact the female athlete's performance. In individuals, dietary needs depend on not only sex, but also size, weight, and energy demands of the sport. A 5-foot female dancer may not require as high a caloric intake as a 6-foot female basketball player. Please see the chapter on nutrition for a more complete discussion of the nutritional needs in athletes in general. This section will focus on vitamin D, calcium, iron, and hydration—areas of particular concern for the female athlete.

Adequate consumption of carbohydrates, protein, fat, and other macronutrients, are critical to replace glycogen stores, and repair exercise-induced tissue damage. On average, an energy

intake of less than 1800 kcal per day in a female athlete can result in a persistent state of negative energy balance and diminish sports performance.

Vitamin D is an emerging micronutrient with deficiency being linked to low bone mineral density (BMD), cancer, heart disease, autoimmune problems, and infections.³¹ Vitamin D is needed for the absorption of calcium, which is critical to bone health. Vitamin D is also important in nervous system and skeletal muscle development and function.³² Female athletes living in northern latitudes who participate in indoor sport are at risk for vitamin D deficiency. Indoor sport athletes are nearly twice as likely as outdoor athletes to be vitamin D deficient.³¹ While sun exposure is important for vitamin D levels, low dietary intake may also lead to deficiency. Vitamin D can be obtained by one of two methods. Endogenous vitamin D is synthesized in the skin following direct sun exposure; exogenous vitamin D is ingested in foods or can be obtained as a supplement. A recent very interesting observation related to vitamin D is the risk of upper respiratory tract infections due to a lowered immune system response in patients with low vitamin D levels.³¹ Respiratory infections are a leading medical cause of lost time for the female athlete. At-risk athletes, especially athletes aged 19 to 49, may benefit from 200 IU of vitamin D supplementation per day.³² Female athletes with the female athlete triad or risk factors for osteoporosis may need 400 to 800 IU per day. (See sections on the [Female Athlete Triad](#) and [Osteoporosis and Osteopenia](#) in this chapter.)

Adequate calcium intake is essential for proper bone mineralization. Females may be more at risk than males for low calcium intake as females more frequently restrict caloric intake and often shun dairy products feeling they are too high in calories. However, this is not the case since we now have options such as low-fat milk, low-fat cottage cheese, low-fat yogurt, and low-fat string cheese that are high in calcium but low in calories. [Table 27.5](#) lists the required daily calcium needs for women of various age ranges. Three servings of daily calcium products typically provide the needed daily requirements. Female athletes should be encouraged to drink low-fat milk for breakfast and eat two servings of low-fat yogurt or string cheese daily to fulfill their

calcium needs. [Table 27.6](#) lists the calcium present in some commonly consumed foods.

Iron is a mineral that has long been known to affect athletic performance in the female athlete. Iron deficiency has been shown to limit endurance training and sports performance.³⁵

TABLE 27.6 Sources of Calcium

Food	Milligrams (mg)	
	Per Serving	Percent DV ^a
Yogurt, plain, low fat, 8 oz	415	42
Mozzarella, part skim, 1.5 oz	333	33
Sardines, canned in oil, with bones, 3 oz	325	33
Yogurt, fruit, low fat, 8 oz	313–384	31–38
Cheddar cheese, 1.5 oz	307	31
Milk, nonfat, 8 oz ^b	299	30
Soy milk, calcium fortified 8 oz	299	30
Milk, reduced fat (2% milk fat) 8 oz	293	29
Milk, buttermilk, low fat, 8 oz	284	28
Milk, whole (3.25% milk fat), 8 oz	276	28
Orange juice, calcium fortified, 6 oz	261	26
Tofu, firm, made with calcium sulfate, $\frac{1}{2}$ cup ^c	253	25
Salmon, pink, canned, solids with bone, 3 oz	181	18
Cottage cheese, 1% milk fat, 1 cup	138	14
Tofu, soft, made with calcium sulfate, $\frac{1}{2}$ cup ^c	138	14
Ready-to-eat cereal, calcium fortified, 1 cup	100–1000	10–100
Frozen yogurt, vanilla, soft serve, $\frac{1}{2}$ cup	103	10
Turnip greens, fresh, boiled, $\frac{1}{2}$ cup	99	10
Kale, fresh, cooked, 1 cup	94	9
Ice cream, vanilla, $\frac{1}{2}$ cup	84	8
Chinese cabbage, bok choy, raw, shredded, 1 cup	74	7
Bread, white, 1 slice	73	7
Pudding, chocolate, ready-to-eat, refrigerated, 4 oz	55	6
Tortilla, corn, ready-to-bake/fry, one 6" diameter	46	5
Tortilla, flour, ready-to-bake/fry, one 6" diameter	32	3
Sour cream, reduced fat, cultured, 2 tablespoons	31	3
Bread, whole wheat, 1 slice	30	3
Kale, raw, chopped, 1 cup	24	2
Broccoli, raw, $\frac{1}{2}$ cup	21	2
Cheese, cream, regular, 1 tablespoon	14	1

^aDVs were developed by the US Food and Drug Administration to help compare the nutrient content within the context of a total daily diet.

^bCalcium content varies slightly by fat content; the more fat, the less calcium the food contains.

^cCalcium content is for tofu processed with a calcium salt. Tofu processed with other salts does not provide significant amounts of calcium.

DV, Daily value.

<https://ods.od.nih.gov/factsheets/Calcium-HealthProfessional/#h2>.

TABLE 27.5 Required Daily Calcium Needs for Women^{33,34,a}

Age Group (years)	Suggested Intake (mg/day)
1–3	700
4–8	1000
9–18	1300
19–50	1000
51–70	1000
>70	1200
Amenorrhoeic athletes ^b (all ages)	1500
Pregnant/lactating women	1500
14–18	1300
19–50	1000

^aRecommendation of National Osteoporosis Foundation (www.NOI.org)

^b<https://ods.od.nih.gov/factsheets/calcium-HealthProfessional>.

Iron is required in hemoglobin and myoglobin production. Since hemoglobin is responsible for carrying oxygen to muscles, adequate levels are vital during exercise, especially for endurance sports. Female athletes are especially at risk for iron deficiency due to menstrual blood loss monthly and poor dietary intake. Vegetarian athletes especially are at risk since red meats are an excellent source of iron. Training at high altitudes, losses of iron in sweat, feces, urine, foot strike hemolysis, and intravascular hemolysis can contribute to iron loss and subsequent deficiency. The female athlete's *iron* is best screened by measuring serum ferritin. The complete blood count is used to screen for *anemia* by measurement of the hemoglobin. The female athlete can have iron deficiency anemia where both the ferritin and hemoglobin are decreased.

Replenishing iron in those who are significantly iron deficient can take 3 to 6 months of dietary changes and supplementation. Therefore diagnosing the deficiency early or identifying at-risk athletes is important in decreasing the potential of loss of performance. Increasing iron by supplementation in the deficient athlete results in increased work capacity by enabling increased oxygen uptake, decreasing lactate concentration, and reducing heart rate during exercise.³² Current recommendations for dietary iron intake are 15 mg/day for girls 14 to 18 years of age and 18 mg/day for women 19 to 50 years of age. The bioavailability of dietary iron can be enhanced by ascorbic acid and fermented foods, and the inclusion of lean meat, chicken, and fish into the diet. Heme iron (type of iron found in animal protein) is better absorbed than nonheme iron, which is particularly important in counseling vegetarian athletes. Examples of iron-rich foods are listed in Table 27.7.

Athletes who are minimally dehydrated can experience increased core temperatures.^{36,37} Females have a higher thermoregulatory threshold than males and begin sweating at a higher core temperature than males. This may make it harder for females to cool during times of intense training.³⁸ Nonreplaced sweat loss of 1% to 2% of body mass can impair performance. One can measure the degree of an individual's hydration by urine-specific gravity or by computing differences between pre- and postexercise weight.³⁹ Water is adequate replacement for fluid

losses unless excessive sweat losses occur as in endurance events. In these instances, adding electrolytes to water or supplying a sport drink with electrolytes is recommended for females just as it is for males.

A sport medicine provider along with a sports' nutritionist can serve as a valuable resource for the female athlete in helping her plan her nutritional needs.

Female Athlete Triad

The female athlete triad is a complex medical syndrome of three interrelated entities: low energy availability (with or without disordered eating), menstrual dysfunction, and low BMD. The latter two entities can be linked to dysfunction secondary to low energy availability (Fig. 27.1).⁴⁰⁻⁴² The female athlete triad appears to be most common in sports where the "lean look" is valued (e.g., running, gymnastics, figure skating, and ballet) but can occur in any physically active female. A female athlete may present with one or more of the three components of the triad and may be mildly to severely affected in any of the three. However, up to 15.9% of athletes with the triad are reported to present severely affected in all three components.⁴³

Energy availability is the term that refers to the total energy needed by a female athlete to perform all of her daily physiologic functions, including athletic activities. Energy availability is defined as energy intake (kcal) minus exercise energy expenditure (kcal) divided by kilograms of fat-free mass or lean body mass.⁴³ Functional hypothalamic amenorrhea can be divided into primary amenorrhea, secondary amenorrhea, or oligomenorrhea. Primary amenorrhea is defined as the absence of menarche by 15 years of age. Secondary amenorrhea is defined as a previously menstruating female having no menstruation for three consecutive menstrual cycles. Oligomenorrhea is a menstrual cycle length greater than 35 days, or less than nine cycles per year. The Z-score measures BMD in premenopausal women. Low BMD (osteopenia) is a Z-score between -1.0 and -2.5, and osteoporosis is a Z-score of less than -2.5 with one or more secondary conditions resulting from low BMD (e.g., stress fractures).

It may be difficult to recognize an athlete with the triad. Presenting symptoms can include disordered eating, hair loss, dry skin, fatigue, weight loss, increased healing time for injuries, increased incidence of stress fractures, and absent menses. The most important screening factors for the triad suggested by the 2014 Athlete Triad Coalition Consensus Statement on Treatment

TABLE 27.7 Examples of Iron-Rich Foods
• Liver • Lean red meats, including beef, pork, lamb • Seafood, such as oysters, clams, tuna, salmon, and shrimp, etc. • Beans, including kidney, lima, navy, black, pinto, soy beans, and lentils • Iron-fortified whole grains, including cereals, breads, rice, and pasta • Greens, including collard greens, kale, mustard greens, spinach, and turnip greens • Vegetables, including broccoli, swiss chard, asparagus, parsley, watercress and Brussels sprout • Chicken and turkey • Blackstrap molasses • Nuts • Egg yolks • Dried fruits, such as raisins, prunes, dates, and apricots • Curry powder, paprika, thyme

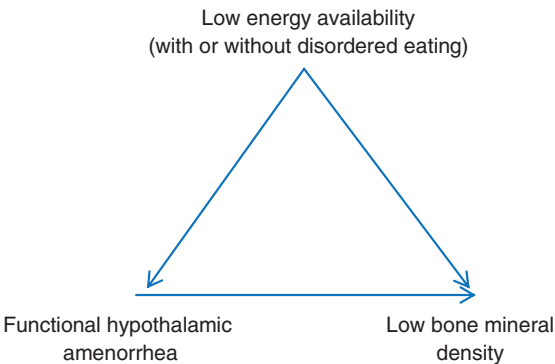


Fig. 27.1 Components of the female athlete triad.

TABLE 27.8 Suggested Screen Questions for Diagnosing the Triad

Have you ever had a menstrual period?
 How old were you when you had your first menstrual period?
 When was your most recent menstrual period?
 How many periods have you had in the past 12 months?
 Are you presently taking any female hormones (estrogen, progesterone, birth control pills)?
 Do you worry about your weight?

*Committee on Obstetric Practice. Physical Activity and Exercise During Pregnancy and the Postpartum Period. No. 650. Washington, DC: The American College of Obstetricians and Gynecologists; 2015:135–142.

DeSouza MJ, Nattiv A, Joy E, et al. 2014 female athlete coalition consensus statement on treatment and return to play of the female athlete triad: 1st international conference held in San Francisco, California, May 2012 and 2nd international conference held in Indianapolis, Indiana, May 2013. *Br J Sports Med.* 2014;48:289–309.

and Return to Play of the Female Athlete Triad expert panel are listed in Table 27.8.⁴¹ Psychologic characteristics that may be present in those with the triad include low self-esteem, depression, and anxiety disorders.^{42,44} Physical findings on exam include anemia, orthostatic hypotension, electrical irregularities, vaginal atrophy, and bradycardia.^{41,45}

The diagnosis of low energy availability can be difficult to make because there is no single test to make this diagnosis. It is recommended that a body mass index (BMI) rather than merely a weight of athlete should be obtained. If the BMI is less than 17.5 kg/m², the state of low energy availability is likely present. Low energy availability has been said to be less than 45 kcal/kg of lean body mass per day. Treatment for low energy availability focuses on a well-designed nutritional plan to attempt to reverse weight loss, achieve a BMI of ≥18.5 kg/m² or 90% of predicted weight, and have a minimum energy intake of 2000 kcal/day.⁴¹ The diet needs to contain a balance of carbohydrates, fats, and proteins. Some athletes have low energy availability not because of inadequate energy intake but because of excessive exercise.

In making the diagnosis of amenorrhea (now known as functional hypothalamic amenorrhea), one must first rule out pregnancy and/or other metabolic or hormonal conditions that can lead to amenorrhea. Some of these conditions include thyroid disorders, hyperprolactinemia, primary ovarian dysfunction, polycystic ovary syndrome, and hypothalamic or pituitary disorders. In the female athlete with primary amenorrhea, physical examination of the female genitalia is mandatory to rule out anatomic abnormalities. Additional testing with blood work and/or hormonal challenges may reveal a correctable metabolic or hormonal irregularity. The treatment for amenorrhea or the menstrual irregularity of the triad is oral contraceptives; however, this is not just to regulate a girl's period but rather to resolve the major issue, which is energy imbalance.

Low BMD can be diagnosed through dual energy x-ray absorptiometry (DEXA) scans. The indications for DEXA scanning include a history of eating disorders, low BMI (≤17.5 kg/m²), low body weight, menarche ≥16 years of age, decreased number of menses over the previous year, history of stress fractures, and

TABLE 27.9 Triad Coalition Panel Recommendations for Dual Energy X-Ray Absorptiometry Scanning**Who Should Get DEXA Scans for BMD Testing**

"High risk" triad risk factors: (need 1)

History of a DSM-V diagnosed ED

BMI ≤17.5 kg/m², <85% estimated weight, OR recent weight loss of ≥10% in 1 month

Menarche ≥16 years of age

Current or history of <6 menses over 12 months

Two prior stress reactions/fractures, one high-risk stress reaction/fracture or a low-energy nontraumatic fracture

Prior Z-score of <−2.0 (after at least 1 year from baseline DEXA)

OR

"Moderate risk" triad risk factors: (need 2)

Current or history of DE for 6 months or greater

BMI between 17.5 and 18.5, <90% estimated weight, OR recent weight loss of 5%–10% in 1 month

Menarche between ages 15 and 16 years

Current or history of 6–8 menses over 12 months

One prior stress reaction/fracture

Prior Z-score between −1.0 and −2.0 (after at least a 1-year interval from baseline DEXA)

BMI, Body mass index; DE, disordered eating; DSM-V, diagnostic and statistical manual of mental disorders; ED, eating disorders.

Joy E, Kussman A, Nattiv A. 2016 update on eating disorders in athletes: a comprehensive narrative review with a focus on clinical assessment and management. *Br J Sports Med.* 2016;50:154–162.

a history of low Z-scores.⁴³ Screening is also indicated if the athlete has two or more moderate risk factors, which are similar to the high risk factors but less extreme; for example, menarche between the ages of 15 and 16 or a BMI of 17.5 and 18.5 or perhaps just one past stress fracture (Table 27.9). Treatment of low BMD should start with increasing energy availability and supplementing with calcium and vitamin D. The use of bisphosphonates, commonly used in postmenopausal females to increase bone density, is controversial and, in fact, is not US Food and Drug Administration approved for the young athlete nor are other medications presently used in treating low BMD in postmenopausal females.

Evaluation and treatment of the female athlete triad is a complex issue and should be determined by a multispecialty treatment team. At present there are no standard guidelines as far as returning the athlete to full participation. One needs to monitor carefully: caloric intake, weight gain, and resumption of menses. Sustained low energy availability can impair health and have associated with it cardiovascular, endocrine, reproductive, skeletal, gastrointestinal, renal, and central nervous system issues. Therefore early recognition and intervention in the female athlete with the triad is essential.

Osteoporosis and Osteopenia

Many women enjoy continuing their athletic careers as they age. There are senior or masters level competitions in a variety of sports. These female athletes exercise and train at a high level of intensity to maintain their sports participation. One of the

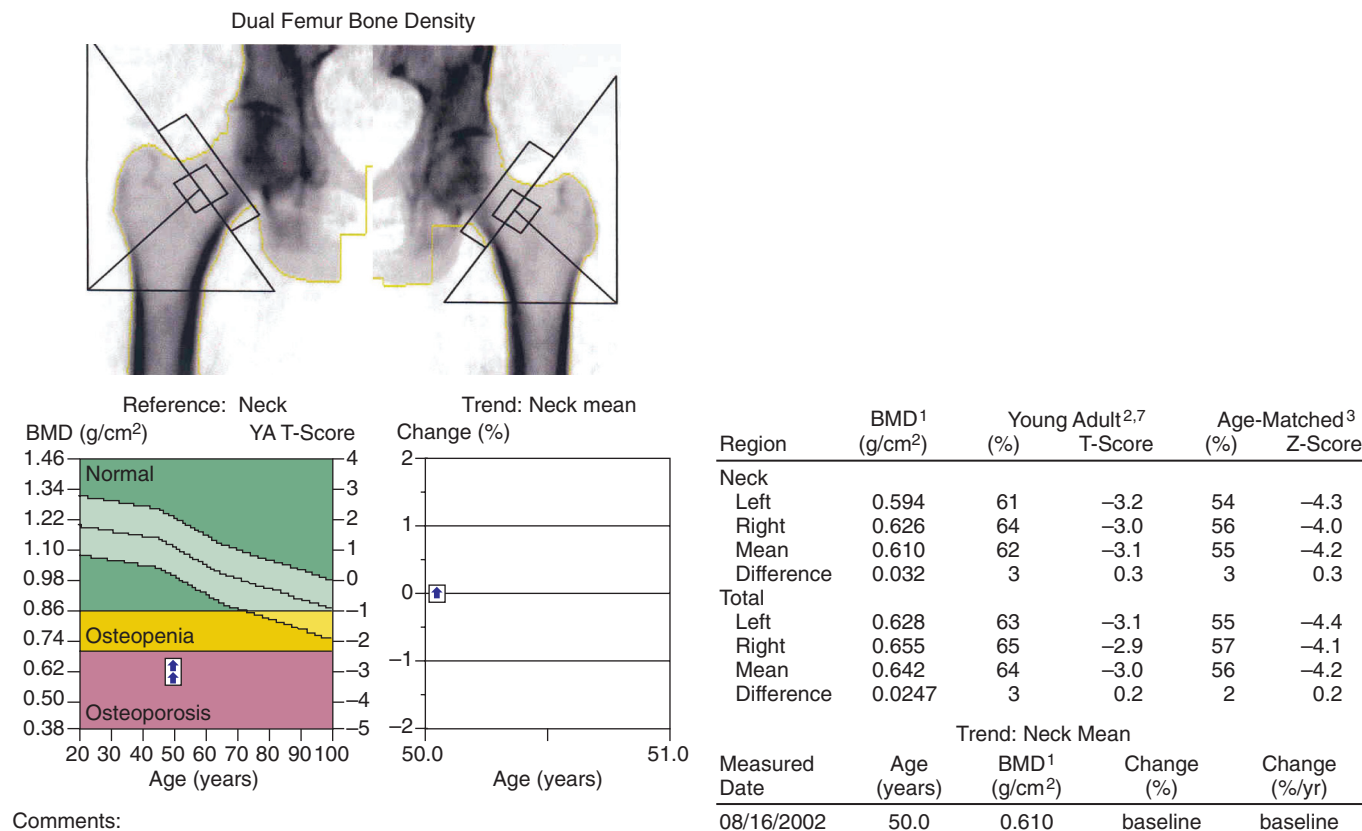


Fig. 27.2 Example of a DEXA scan.

TABLE 27.10 Defining Osteoporosis and Osteopenia Using the WHO “T-Score”

Normal BMD	BMD within 1 SD (T-score: +1 or -1) of the young adult mean
Osteopenia	BMD between 1 and 2.5 SD below the young adult mean (T-score: -1 to -2.5 SD)
Osteoporosis	BMD 2.5 SD or more below the young adult mean (T-score: -2.5 SD or lower)

BMD, Bone mineral density; SD, standard deviation; WHO, World Health Organization.
www.nof.org.

consequences of menopause in a female is a decrease in bone density secondary to decreasing estrogen levels, which results in an increased risk of fractures. As noted previously in the Female Athlete Triad section, BMD is most reliably measured by DEXA scanning (Fig. 27.2). Scores obtained are compared with a healthy, average, young adult population to yield a *T-score* rather than a *Z-score*, which is most frequently used in a younger population to compare the athlete’s bone density to someone of like age. According to the World Health Organization, osteopenia is defined as a *T-score* of -1.0 to -2.5 standard deviation (SD) below normal average age-matched BMD, and osteoporosis is a *T-score* lower than -2.5 SD below average age-matched BMD, as measured by the DEXA scan Table 27.10.

There are specific indications for DEXA testing in the post-menopausal female: females over the age of 65, or females who

are 50 to 65 who have a suspected secondary cause of osteoporosis such as hyperthyroidism, inflammatory bowel disease, chemotherapy, and hyperparathyroidism. Females 60 to 64 should be screened if they have a history of alcoholism, tobacco use, low weight (<125 pounds), sedentary lifestyle, or estrogen deficiency from a premature menopause either natural or as a result of hysterectomy and oophorectomy. Other risk factors that may merit early DEXA testing are first-degree relatives with osteoporotic fractures, previous history of fracture over 40 years of age, Caucasian or Asian heritage, and an anovulatory state for more than 5 years on no estrogen replacement.

Treatment for bone loss should start early in the osteopenia phase, if not before, to potentially slow the rate of the loss. Treatments for osteopenia and osteoporosis are both nonpharmacologic and pharmacologic. The primary nonpharmacologic treatment is weight-bearing exercise. Females should also stop smoking and avoid excessive weight loss. BMD is higher in postmenopausal females who exercise.⁴⁶ It has been shown that athletes (runners) who performed weight-bearing exercise had a higher BMD than athletes (swimmers) who did non-weight-bearing exercise. The athletes in this study had a higher BMD of the femoral neck than sedentary women.⁴⁶ Pharmacologic treatment includes 1000 to 1500 mg of calcium and 400 to 800 IU of vitamin D supplementation for at-risk females. The most controversial treatment may be hormone replacement therapy. The US Preventive Services Task Force (USPSTF)⁴⁷ along with the American Academy of Family Physicians (AAFP), and other groups, recommend against menopausal hormone therapy to prevent chronic conditions.

This includes a combination of estrogen and progestin, or estrogen therapy alone.⁴⁷ The best treatment for the female athlete with osteopenia and osteoporosis is not clear. It is known that weight-bearing exercise, calcium and vitamin D supplementation and smoking cessation are important. Athletes should consult their health care provider to make an informed decision as to what may be best for them.

Birth Control

There are many contraceptive choices available to women. Between 2006 and 2008, 7.3% of women who were at risk for an unintended pregnancy were not using a contraceptive method.⁴⁹ The female athlete, if sexually active, should have a yearly gynecologic examination and review contraceptive choices with her physician. Multiple alternatives for contraceptive use are available and include: intrauterine devices, spermicidal products, diaphragms, cervical caps, female condoms, male condoms, injectable progestin, vaginal ring (NuvaRing), implantable hormonal inserts, and oral contraceptives (Table 27.11). See Table 27.12 for a review of major risks and benefits of each of these methods of contraception.

At present, the most popular form of contraception is the oral contraceptive pill (OCP). Two types of OCPs are the progestin-only pill and the combination estrogen and progestin pill. The OCPs are 91% to 99% effective in preventing pregnancy. Their mechanism of action is the exogenous hormone in the pill, which suppresses the pituitary to decrease production of follicle-stimulating hormones and luteinizing hormones and thus suppresses ovulation. Female athletes who suffer from heavy or painful menstrual cycles may get decreased flow and pain relief from the OCP. This use may be the most important

noncontraceptive use for the OCP. Contraindications to the use of the OCP include personal history of blood clots, abnormal liver functions, breast cancer, migraines, and certain cardiac conditions. Many female athletes prefer not to take the OCP out of fear of weight gain. Several excellent studies have shown that OCPs are not responsible for weight gain, and in a study on runners compared to nonathletic controls, OCPs had neither an increase of weight, nor an increase of fat mass or body fat compared to nonathletic controls.^{50,51}

A female's choice to start contraception and the method used should solely be hers. The goal of the sports medicine provider should be to provide accurate information about each method and to address concerns or questions to help the female athlete find a safe and effective method of birth control that does not interfere with her athletic lifestyle or put her health at risk.

Pregnancy

Many of the early exercise guidelines during pregnancy were based on supposition rather than well-executed scientific studies. Previous concerns that exercise might result in preterm labor are unproven.⁵² It is currently recognized that mild to moderate exercise during pregnancy is not only safe but advantageous.⁵³ In 2015, the International Olympic Committee organized a meeting of 15 experts on the physiologic effects of exercise during pregnancy. Consensus from this meeting was that while pregnant women should be encouraged to "maintain an active lifestyle, i.e. do mild to moderate exercise, the limits and intensity of highly strenuous exercise on the fetus has not been thoroughly researched."⁵⁴

Benefits of regular exercise include: improving self-image, avoiding excessive weight gain during pregnancy, aiding in the management of gestational diabetes, decreasing musculoskeletal complaints of pregnancy (i.e., an increase in core strength can result in fewer complaints of low back pain). In order to ensure safety, the pregnant female should undergo a preexercise program evaluation with her obstetric provider to review her desired program in terms of both the actual activities and intensity, and the length of the workout. Contraindications to exercise during pregnancy are listed in Table 27.13. Sports that are safe for the pregnant athlete and sports to avoid during pregnancy are listed

TABLE 27.11 Classes of Contraceptive Methods

IUD	Progestin
Spermicidal products	Vaginal ring
Condoms	Implantable hormones
Diaphragms	Oral contraceptives

IUD, Intrauterine device.

TABLE 27.12 Risk and Benefits of Popular Contraceptive Methods

Contraceptive Method	Risks	Benefits
Intrauterine device	Perforation of the uterus	Length of time it can be used
Spermicidal products	Used alone, low effectiveness	Improves effectiveness in preventing pregnancy when used with diaphragm, cap, condoms
Diaphragm	Low effectiveness, user error	Nonhormone contraception
Cervical cap	Low effectiveness, user error	Nonhormonal contraceptive
Condoms (male/female)	Allergic reaction to latex	Low cost, may reduce incidence of STI
Injectable progestin	Variation of menstrual bleeding	Only has to be administered every 3 months
	Injection site reaction	
Vaginal ring	Blood clots from hormone	Does not need to be taken or changed daily
Implantable hormonal inserts	Skin infection at insertion site	Long-term use (3 years)
Oral contraceptives	Blood clots, migraines	Most effective method if used correctly

STI, Sexually transmitted infection.

TABLE 27.13 Contraindications to Exercise During Pregnancy^{55,a}

Relative
Poorly controlled maternal hypertension
Poorly controlled maternal diabetes
Previous spontaneous abortion
Severe anemia
Underweight or maternal eating disorder
Absolute
Preeclampsia/eclampsia
Premature rupture of membranes
Placenta previa
Persistent 2nd or 3rd trimester bleeding
Preterm labor
Incompetent cervix
Growth restricted fetus

Committee on Obstetric Practice. *Physical Activity and Exercise During Pregnancy and the Postpartum Period*. No 650. Washington, DC: The American College of Obstetricians and gynecologists; 2015:135–142.

TABLE 27.14 Safe Sports and Sports to Avoid When Pregnant

Safe Sports During Pregnancy^a	
Walk	Aerobics
Swim	Pilates
Cycle	Yoga
Run	
Sports to Avoid During Pregnancy	
Contact sports can potentially cause harm to the unborn fetus.	
Downhill skiing includes the risk of fall and altitude sickness.	
Scuba diving has the potential for decompression sickness.	

^aAlways check with your physician first.

in Table 27.14 A healthy pregnancy and delivery should be the ultimate goal, and a well-structured exercise program can help attain that goal.

Several exercise precautions are recommended. The pregnant female needs to avoid substantial or sudden increases in temperature, as an increase in core body temperature greater than 103°F can result in neural tube defects during the development of the neural tube.⁵⁶ Pregnant females should avoid exercise in extreme heat and humidity, and should workout early in the morning or later in the day when temperatures are cooler, decrease the intensity of training (especially in hot environments), and ensure adequate fluid intake.

Postural changes (progressive lumbar lordosis and anterior rotation of the pelvis), which result from the pregnant uterus moving the center gravity forward, can combine with increased size and weight of breast tissue to necessitate significant balance adaptations to avoid injury while exercising. Signs that cause the pregnant mother to refrain from or discontinue exercise instantly including vaginal bleeding, amniotic fluid leakage, decreased fetal movement, chest pain, shortness of breath (before and during exercise), and dizziness.

The National Collegiate Athletic Association (NCAA) has guidelines for the pregnant student athlete that focus on the athlete’s safety, and the safety of the unborn fetus.⁵⁷ The athlete can be granted a 1-year extension of the 5-year period of eligibility to account for lost time during the pregnancy. The NCAA guideline on pregnancy states that the team physician and the student’s health care provider should work together to help the pregnant student athlete safely continue her sport participation. In any differences of opinion regarding the pregnant student athlete’s medical care, the team physician should defer to the student athlete’s health care provider.⁵⁷ The institution does not dictate the medical care.

Strenuous exercise can be started 6 to 8 weeks after delivery. Exercise does not have an adverse effect on lactation, and neither the volume nor composition of breast milk.⁵⁸ Thus a new mother can continue or resume her exercise program without worry and breastfeed without fear of negatively affecting the numerous benefits of breastfeeding.

Psychological Issues in the Female Athlete

It is important to encourage adolescent girls to participate in sport since it has been shown to increase self-esteem and decrease the incidence of depression.⁵⁹ Sport-active teenage girls are less likely to get pregnant and are more likely to graduate from high school.⁵⁹ They are also less likely to take drugs during their teenage years. The Women’s Sport Foundation also emphasizes that sport participation in women will decrease shyness, increase social skills, decrease social anxiety, and help young females cope better with success and failure. Father/daughter bonding frequently develops through sport participation.^{59–65}

Opinions are varied regarding the development of muscular bodies in females who exercise, from a perception of “increased sexuality” to “less feminine.”^{61,64} Years ago Goffman wrote that body image is a combination of many factors including the selection of clothing, the manner of carriage, and speech.⁶⁶ As females pursue characteristics associated with successful sporting performances (e.g., competitiveness and independence), and as they develop bodies that are stronger and more muscular, they may have difficulty accepting these characteristics that were once thought to be “feminine-threatening” characteristics. Over the last 30 years, the muscular body of the exercising female has become more desirable. The antithesis to the sports where the “thicker” muscular physique is desirable are sports where the lean look is favored (e.g., cross country, figure skating, and ballet). Girls involved in these activities are at risk for eating disorders as they may have an altered body image and perceive themselves as always being “too heavy.”⁶⁷ Those involved in women’s sports must recognize and be sensitive to these body image dilemmas that the young athletic female may face.

Moreover, most coaches of junior high, high school, or collegiate sports will tell you that coaching women necessitates a different approach than that used when coaching men. Although admittedly a generality, male players usually will respond to intimidation and encouragement to demolish the opposition whereas these tactics are typically not effective for female athletes. It is more effective to appeal to a female’s emotions and encourage her to work harder out of loyalty to her team and coaches.

Boys and young men believe that proof of masculinity is rooted in sport participation; while girls and young women typically receive little such secondary gain from sports. In fact, girls—even if they are superb players—have been known to drop out of a sport to explore other activities. Coaches and parents may find this sport “burn out” or change of activity focus challenging to accept. The young athlete may find it difficult to explain to parents and her coach that she no longer wishes to participate in her sport and may find it more “socially acceptable” to have an “injury” that sidelines her. In this scenario, it is the health care provider’s responsibility to be cognizant of the fact that an injury may merely be the superficial expression of a deeper issue, and they can help the young athlete and her family to recognize this reality. To retain female players during the teenage years, coaches often must be creative to make their sport enjoyable as well as competitive. As females progress through higher levels of sports their desire and motivation become more similar to their male counterparts.

Degenerative Arthritis and the Female Athlete

An important question frequently posed by the athletic female is, “Will exercise improve symptoms of osteoarthritis or does regular exercise enhance the risk of osteoarthritis?” In the United States, 27 million individuals are reported to have osteoarthritis and in the population older than 45 years of age, osteoarthritis is more common in females than males.⁶⁸ Osteoarthritis occurs both in athletic individuals and in those who are sedentary. Genetics is the largest risk factor for osteoarthritis and in those with knee arthritis (the most frequent presentation of osteoarthritis), obesity is the second greatest risk factor, and being female is the third largest risk factor, ranking higher than prior trauma.⁶⁹

In the past, when athletes developed arthritis, they were often discouraged from exercising. On the contrary, the recommendation of Healthy People 2010 included educating individuals with arthritis about their disease process, the significance of obesity as a risk factor for osteoarthritis, and the value of maintenance of an exercise program.^{68,70} Recommendations for exercise include strengthening programs, stretching, and balance and agility drills. In some individuals with significant degenerative changes in lower extremity joints, decreasing weight-bearing exercise while increasing non-weight-bearing exercise (swimming, water exercise, biking, use of the elliptical machine, or using light weights with multiple repetitions) is recommended.^{70,71} Water exercise is effective in those with severe arthritis who find impact activity painful, but many with significant arthritis of the spine and lower extremities can do land exercise to increase mobility and decrease the pain of arthritis while also improving balance and agility.⁷² The premise is that strong muscles help protect arthritic joints, which is especially true with lower extremity osteoarthritis where a strong core and strong lower extremity muscles can help to unload an arthritic joint.

Moreover, exercise has been found to improve self-esteem and decrease depression.^{73,74} Flexibility exercises decrease stiffness. The combination of enhancing balance and decreasing stiffness helps to minimize the chance of falls in the arthritic population.⁷⁵ Furthermore, Kushi reported an association between increased physical activity and decreased risk of mortality.⁷⁶

Exercise also helps to maintain or decrease weight. Since obesity is associated with an increased risk of developing arthritis, controlling weight helps decrease the incidence of osteoarthritis.⁷⁷ Curl reported that the risk of cardiovascular disease across all age groups in females decreased as a result of increased fitness.⁷⁴

Moreover, studies by Hannan and Panush as well as Chakravarty have demonstrated that the incidence of osteoarthritis is not increased in an exercising population when compared to age-matched controls.^{78–80} Although sports that subject joints to repetitive high levels of impact and horizontal loading may increase the risk of articular cartilage degeneration, moderate exercise has not been found to increase the risk of osteoarthritis.^{80,81}

Concussions

A concussion is defined as a traumatic brain injury resulting in a transient disturbance of brain function. The injury results when linear and/or rotational forces affect the brain. The process involves a complex pathophysiologic process, which involves both ionic and metabolic components at the microscopic axonal level.⁸² Research continues to study changes that occur in brain function acutely and the effects that such changes have on the long-term function of the brain. The hope is to potentially improve outcome by lowering the risk for progressive and degenerative changes seen in athletes who have suffered multiple concussions.

The female athlete with a suspected concussion needs to be evaluated acutely and carefully in a controlled setting. Depending on the mechanism of injury, cervical spine injuries need to be excluded. The female athlete should not be allowed to return to play (RTP) on the day of their injury. Components of the concussion evaluation need to include obtaining subjective symptoms, evaluating the physical signs, the behavioral changes, and any cognitive impairment.⁸³ The severity of subjective symptoms, such as headache, feeling in a fog, and emotional symptoms, needs to be obtained. Included in the initial evaluation is the determination of the loss of consciousness and amnesia, both anterograde amnesia and retrograde amnesia. Anterograde amnesia involves memories after the injury and retrograde involves loss of memories before the injury. Numerous quick, concise, and organized sideline assessment tools have been developed to assist with evaluation at the time of the injury. The SCAT 5⁸⁴ sideline assessment tool, and its variations, is commonly used for sideline evaluation, but other tools do exist.⁸³ The best tool is the one the clinician feels the most competent using. A more complete evaluation in a training room, medical office, or emergency room needs to be completed after the initial assessment. This includes a comprehensive physical exam including a comprehensive neurologic exam with attention paid to motor, sensory, balance, and cognitive components. If there is a concern for an intracranial hemorrhage an emergent noncontrast computed tomography scan is ordered.

Once the acute assessment has been completed the management and treatment phase begins. It is widely accepted that physical and mental rest is the most important initial phase of the treatment. Neuropsychological testing are objective tests used to assess cognitive impairment. The tests may be administered with paper and pencil or by a computerized version. The best type of test and the timing of the test have not been

TABLE 27.15 Example of Return-to-Play Protocol Following a Concussion

The athlete must perform each activity without symptoms before progressing to the next phase
No activity, physical and mental rest
Light aerobic activity
Sport-specific exercise
Noncontact training drills
Full contact practice
Return to game play

clearly determined. It is widely accepted that the tests need to be interpreted by trained individuals, such as a neuropsychologist, with experience using the testing tool. Some clinicians utilize neuropsychological testing during various stages in the athlete’s RTP protocol. When the athlete becomes asymptomatic they may start a RTP protocol, which initially begins with light aerobic exercise eventually progressing to a full contact practice. If the female athlete progresses through each phase without a return of symptoms they can be returned to full play (Table 27.15).⁸³

Concussions affect males and females differently, and evidence indicates that female athletes may be at greater risk for concussions than male athletes. Analyses of injury data from the NCAA databank reveal that compared to males, female athletes sustain more concussion in games than in practice.⁸⁵ Female soccer players have the most concussions followed by female basketball, lacrosse, softball, and gymnasts. Female soccer and basketball players sustain significantly more concussion than their male counterparts in their respective sports.⁸⁵ This trend was recently confirmed in a study involving high school athletes. In gender comparable sports (soccer, basketball, and baseball/softball), high school girls had higher rates of concussions (new and recurrent concussions) compared to high school boys and concussion represented a greater proportion of all injuries.^{86,87} Concussions in high school girls are more frequent from contact with playing surfaces compared to boys, whose concussions occur following player to player contact.⁸⁶

In addition to a higher number of concussions in certain sports, concussions in female athletes tend to be more severe. A meta-analysis review concluded that traumatic brain injury results in more somatic symptoms in females than in males, including poor memory, dizziness, fatigue, irritability in response to light and noise, impaired concentration, headache, anxiety, and depression.⁸⁸ Additionally, concussed female athletes are reported to be more cognitively impaired than males and have demonstrated significantly lower visual memory composite scores compared to males.⁸⁸ Without the use of helmets, female athletes are twice as likely than males to experience cognitive impairment following concussions.⁸⁹

A recent study has also suggested that female athletes may take 6 days longer to begin their RTP protocol after a concussion compared to age-matched male controls.⁹⁰ In a study of US collegiate athletes’ neuropsychological testing, it was found that female athletes performed significantly better on verbal memory

compared to male athletes, whereas male athletes performed significantly better on baseline visual memory.⁹¹

Compared to new concussions, recurrent concussions tend to be more serious with delayed symptom resolution and “lost athletic time.” Recurrent concussions also have the potential for unfavorable long-term aftermaths. In the high school population involved in “gender comparable sports,” although girls had higher rates of recurrent concussions, the number was not statistically significant, which may indicate that, in this age group, gender differences equalize after the initial concussion.⁸⁶ Some concussed athletes suffer from lingering postconcussive symptoms including chronic headaches. Female athletes report a greater incidence of symptoms from postconcussion syndrome and depression than male athletes.⁸⁹

Various theories have been proposed to explain the differences between female and male concussions. Suggested mechanisms include biomechanical differences in head and neck acceleration forces, hormonal variances, or a perception that boys are “tougher” than girls and continue to play through injury.⁸⁶

ORTHOPAEDIC INJURIES IN THE FEMALE ATHLETE

Epidemiology of Sports Injuries

Much has been written on the epidemiology of sports injury. When one understands the incidence of sports injuries, one can then investigate the risk factors for the most common injuries or injuries associated with the greatest time loss from sport, and by decreasing these risks, can then increase the safety of sport. Although the majority of articles on the differences in sports injuries between males and females limit their discussion to a single sport, some articles do discuss differences between injuries in male and female athletes across a variety of sports often for a single age group (pediatric, adolescent, collegiate, or postcollegiate age athlete).

An epidemiologic study on pediatric sport injuries concluded that in young males the injuries differ by injury type, diagnosis, and body area from those of young females. A retrospective chart review of a 5% random sampling of medical records of children aged 5 to 17 years seen over a 10-year period in a sports medicine clinic in a large academic pediatric hospital (Boston Children’s) concluded that female athletes had a higher percentage of overuse injuries compared to traumatic injuries (62.5% vs. 37.5%), whereas male athletes had a higher percentage of acute injuries (58.2% vs. 41.9%). Females sustained more injuries to the lower extremity (65.8% vs. 53.7%) and spine (11.3% vs. 8.2%), and males sustained a higher percentage of upper extremity injuries than females (29.8% vs. 15.1%). With regard to hip and pelvis areas, females had more overuse and soft tissue injuries and males had the greater number of traumatic bony injuries. Anterior (patellofemoral) knee pain was reported to be three times greater in the female pediatric population than in the male population. Notable also in this group was that ACL injuries were almost equal in the pediatric female and male populations.⁹²

In a 20-year study of multisport middle school injuries, girls exhibited a higher rate for all injuries and a higher rate for time loss than boys in matched sport. Girls had a higher injury rate

during practices than during games for all sports. Practice and game injury rates were nearly identical for boys in all sports. In this study, injuries to middle school athletes in this study were less frequent and less severe than those reported for secondary school and college athletes.⁹³

A study from Kaiser Permanente in California compared the pattern of injuries in male versus female athletes in seven college sports (basketball, cross country, soccer, swimming, tennis, track and field, and water polo) in a Division III college during the years 1980 to 1995. A total of 1874 sport-related injuries occurred during this period, 856 (45.7%) in females and 1018 (54.3%) in males. Female swimmers had more back/neck, shoulder, hip, knee, and foot injuries than their male counterparts, and female water polo players reported more shoulder injuries. In summarizing injuries in all sports, females reported a higher rate of hip, lower leg, and shoulder injuries while males reported a higher rate of thigh injuries. However, on the whole, the authors stated that their data suggest very little difference in the pattern of injuries between males and females competing in comparable sports.⁹⁴

A study of 573 Division I athletes participating in 16 sports over three seasons identified 1317 reported injuries. Female athletes had a lower acute injury rate than male athletes (38.6 vs. 49.8 per 1000 athletic exposures). Males had less overuse injuries than did females (13.2 vs. 24.6 per 1000 athletic exposures).⁹⁵ In a study of 16 collegiate sports over 6 years and 14 high school sports over 7 years, Roos et al. also reported a higher rate of overuse injuries in females compared to males in sex comparable sports at both the high school and collegiate level.⁹⁶ In an analysis of injury rates for time loss and non-time-loss injuries among college student athletes Powell and Dompier found that non-time-loss injury rates were 3.5 times the time-loss injury rate for males and 5.1 times the time-loss injury rate for females.⁹⁷

Foot and Ankle Injuries

Lateral ankle sprains are common injuries in both male and female athletes. However, minor ankle sprains are 25% more common in female basketball players than in male basketball players although the incidences of severe ankle injuries appear to be equally divided between males and females.⁹⁸ The increased ankle ligamentous laxity in females has been implicated as a risk factor for the disparity of minor ankle injuries in the female.⁹⁹ Interestingly, in pro basketball athletes, males and females have similar rates of injury.^{100,101} However, a study of cadets at the United States Military Academy revealed that male cadets had a greater risk of medial ankle sprains than female cadets. Although the risk of syndesmotic sprains was not statistically different for male or female cadets, females with syndesmotic sprains of their ankle had a higher time loss to injury.¹⁰²

Achilles tendon traumatic ruptures have been reported to be more common in males, and males are reported to have better outcomes at 1 year than females regardless of the treatment method.^{103,104} Posterior tibial tendon dysfunction is, in general, more common in older female athletes than in all male athletes, but the risk factors that make this overuse entity more common in females have not been clearly identified, with the exception of obesity.¹⁰⁵ Shin splints appear to be more common in females

TABLE 27.16 Foot Conditions More Common in Females Than Males

Bunions	Hammer toes
Bunionettes	Digital neuromas
Plantar fasciitis	Metatarsal stress fractures

perhaps related to the greater incidence of pronation in the female.¹⁰⁶

Perceived functional limitation from foot pain is more common in female than male athletes and nonathletes.¹⁰⁷ Females are more commonly affected by bunions, bunionettes, plantar fasciitis, hammer toes, digital neuromas, and stress fractures of the metatarsals (Table 27.16). Even in young females (less than 18 years old) foot pain is common (9%) although not as common as in older females where the incidence of foot pain is reported to be 75%.¹⁰⁸

The shape of a woman's foot and the proportion of foot size and shape to body size and shape is different in females than in males.¹⁰⁹ Males have a greater foot length compared to body height than do females.¹¹⁰ Moreover, females have a 20% to 25% lower volume and lower surface area in their subtalar ankle joints with thinner cartilage in these areas as well.¹¹¹ Females have a narrower heel compared to forefoot width than do males. In fact, overall, the female foot is smaller than the male foot compared to body height.¹⁰⁹ Therefore a woman's shoe, especially her athletic shoe, should be made not as a "smaller male shoe" but rather different from a man's shoe in all the parameters noted above.¹⁰⁹

Females' feet differ from males not only anatomically but also functionally. For example, females pronate more than males, which may be related to women's greater tendency to increased joint laxity.¹⁰⁶ Also, female runners have been found to be predominantly midfoot strikers,^{106,112,113} have greater foot abduction with running and greater rear foot motion than male runners.¹¹³

Since the rate of foot problems in populations that do not wear shoes has been historically reported to be low,¹¹⁴ shoes have been implicated as the cause of many of the foot ailments in females including bunions, bunionettes, corns, and hammer toes. Indeed, it has been reported that 8% of females wear shoes smaller than their feet and that this trend starts with adolescence.

Moreover, as females age, their feet become stiffer and flatter. Shoe size increases. Proper shoe fit is essential for avoiding foot issues. It has been suggested that females should buy shoes later in the day as feet often swell during the day, especially with prolonged standing. Girls and women should remember that athletic shoes should "fit" when they are purchased. Buying smaller, improperly fitted shoes and hoping the shoes will "stretch out" over time is highly discouraged as such a practice can lead to foot issues. Remember the adage, "If the shoe fits wear it!" or in this case, if the shoe doesn't fit, don't buy it!

Treatment of most forefoot abnormalities in female athletes is initially conservative care with taping, pads, creams, moleskin, foot and toe exercises, and shoe modification. Surgery should be approached with great caution as secondary biomechanical problems can result. Fig. 27.3 is a radiograph of a young

cross-country runner who had bunion surgery and then developed stress fractures in the second and third metatarsals from altered mechanical forces resulting from a shorter first metatarsal.

Patellofemoral Pain Syndrome

Patellofemoral syndrome (PFS) is one of the most common causes of anterior knee pain with symptoms affecting nearly 30% of the adolescent population.^{115,116} The incidence of PFS is estimated at 22/1000 persons per year.^{117,118} The prevalence of PFS in adolescent athletes is 20% with females affected at a rate of 2 to 10 times higher than their male counterparts.¹¹⁹⁻¹²¹

Patellofemoral pain syndrome (PFPS) manifests as diffuse pain in the anterior aspect of the knee, which is increased by squatting, kneeling, running, and ascending or descending stairs. The onset is insidious and is provoked by overuse or underlying malalignment. New exercises or increases in frequency, intensity, or duration of sports activities are historical facts that suggest an overuse etiology. Exercises, such as squats, lunges, or leg extensions, can increase forces up to three to four times the body weight across the patellofemoral joint,¹²²⁻¹²⁴ and are commonly an inciting factor.

The patella lies proximal to the trochlea when the knee is in full extension and will engage the articulation as knee flexion

approaches 20 degrees, the patella is constrained within the trochlea, by the interaction of bony and soft tissue restrictions. Soft tissue restraints are most important in early flexion, yet as knee flexion increases, the congruency of the bony articulation becomes more pivotal as the patella is steered into the trochlea. Soft tissue constraints primarily consist of the quadriceps and patellar tendon, medial patellofemoral ligament, medial patellar retinaculum, and medial patellomeniscal ligament. Insertions of the vastus lateralis and vastus medialis oblique (VMO) are particularly important because contraction of these muscles can vary the “line of pull” on the patella. Deconditioning, altered activation, or tightness of these anatomic constructs can lead to an alteration in patellofemoral joint mechanics, ultimately resulting in abnormal patellar tracking and the development of pain.

Anterior knee pain is a common presenting symptom, and before making a diagnosis of PFPS patients should be evaluated thoroughly for other underlying pathology, such as patellar instability, osteochondral deficits, plica, and overuse injuries to the extensor mechanism, as all of these can cause pain in the anterior aspect of the knee (Table 27.17). Inciting and alleviating factors, a history of trauma, sports participation, medical history, and previous treatment are important to note. A history of patellar dislocations can help differentiate pain from instability and abnormal patellar tracking. Recurrent effusions, mechanical symptoms, and grinding should be documented and are more likely to indicate an underlying cartilage problem. Physical examination should focus on the entire lower extremity and not solely on the knee joint. Lower extremity alignment, quadriceps (Q) angle, femoral anteversion, tuberosulcus angle, and heel alignment should be assessed. Gait abnormalities, such as in-toeing, hip abductor weakness, hip and knee external rotation, and contractures of the hip abductors should be noted. The Q angle should be examined in full extension and 30 degrees of flexion because a laterally deviated patella in full knee extension can falsely decrease the Q angle. Quadriceps atrophy, especially of the VMO, is common in persons with chronic PFPS. Patellar mobility, tilt, and subluxation should be assessed both medially and laterally. A sensation of “catching” with patellar mobilization against the trochlea is suggestive of cartilage defects. The J sign suggests attenuations of the medial soft tissues and is performed by having a patient slowly extend the knee from full flexion. A positive sign is verified by lateral patellar subluxation as the knee



Fig. 27.3 Stress fracture of the second metatarsal in a cross-country runner following bunion surgery.

TABLE 27.17 Differential Diagnoses of Anterior Knee Pain		
Skeletally Immature Patients	Skeletally Mature Patients	All Patients
Osgood-Schlatter disease	Patellofemoral osteoarthritis	Patellar instability
Sinding-Larsen-Johansson syndrome	Posttraumatic arthritis	Articular cartilage injury
Distal femoral bone tumor	Patellar stress fractures	Chondromalacia patella
Osteochondritis dissecans	Quadriceps tendinopathy/rupture	Loose bodies
Symptomatic bipartite patella	Patellar tendinopathy/rupture	Inflammatory arthritis
Referred pain (Legg-Calvé-Perthes/ Slipped capital femoral epiphysis)	Referred pain (hip arthritis/lumbar spine pathology)	Infectious arthritis/bursitis
	Hoffa disease	Complex regional pain syndrome
		Iliotibial band syndrome
		Bursitis (pes anserine/prepatellar)
		Plica syndrome

approaches extension and leaves the constraints of the trochlear groove.

In the absence of structural disorders, the development of patellofemoral pain is multifactorial. In a systematic review, Lankhorst et al. pooled data from 47 studies and examined 523 variables.¹²⁵ Compared with control subjects, persons with PFPS demonstrated larger Q angles, less hip abduction strength, lower knee extension peak torque, and less hip external rotation strength.¹²⁶ Alterations in neuromuscular activation patterns have also been reported. Electromyography studies have found delayed activation of the VMO compared with the vastus lateralis in patients with PFPS.^{127,128} Multiple retrospective studies have reported that PFPS is associated with weak hip abduction. However, these conclusions have been called into question. The retrospective nature of the studies begs the question whether muscle weakness is a cause or an effect of PFPS. Three recent prospective studies suggest that female athletes with greater hip abduction strength may be at an increased risk for the development of PFPS.¹²⁹⁻¹³¹ Increased cartilage stress may play a role in the development of PFPS. Farrokhi et al. examined patellofemoral biomechanical factors and found that compared with pain-free control subjects, women with patellofemoral pain demonstrated elevated hydrostatic pressure and shear stress across the patellofemoral joint.¹³² Carlson et al. evaluated patellar tracking in female athletes and found that rigorous athletic training might not be adequate to produce PFPS in female adolescents; rather, it may be caused by a combination of physical activity and pathologic kinematics.¹³³

Treatment for PFPS is conservative and revolves around physical therapy. The goal of physical therapy is to restore soft tissue muscular and capsuloligamentous balance of the patellofemoral articulation. Rehabilitative exercises should include a quadriceps, hamstrings, and iliotibial band stretching regimen to restore flexibility, as well as patellar mobilizations to relax contracted lateral capsular structures, quadriceps, and the patellar tendon. A strengthening program that involves the hip abductors and external rotators as well as the quadriceps should be instituted. It is important to protect patellofemoral articulation from high stress by focusing on isometric and short arc closed chain concentric and eccentric muscle strengthening and avoiding open chain exercises and deep flexion weight-bearing activities. McConnell patellar taping, kinesio-taping or patellar bracing to centralize the patella also may be useful during sports participation.

Anterior Cruciate Ligament Injuries

With an exponential rise in the number of females participating in organized athletics, physicians are treating an increasing number of ACL tears in this population. Several studies have demonstrated an injury rate of 1 to 3 per 10,000 athletic exposures in females,¹³⁴⁻¹³⁷ which is two to six times the rate seen in males.^{100,135,138-140,142} Most of these ruptures occur during pivoting, cutting, changing directions, and jumping.^{136,138,143-148} This disproportionate injury rate has been the driving force for numerous studies on noncontact ACL injuries in females including risk factors and prevention programs for vulnerable athletes.^{136,138,139,142,149-154}

Risk Factors for Noncontact Anterior Cruciate Ligament Injury

Sex differences in the rate of noncontact ACL injury are multifactorial. Any factor that either increases the load placed on the ligament, or lowers the ligament's ultimate load to failure will increase the rate of rupture.¹⁵⁰ Potential risk factors can be grouped into anatomic, hormonal, environmental, and neuromuscular factors (Table 27.18).

Anatomic risk factors for ACL injury in female athletes are body mass, ligament size and mechanical properties, bony morphology, and limb alignment. A high BMI has been postulated to decrease core stability, resulting in excessive trunk motion and an increase in valgus load on the knee. For these reasons, BMI is associated with increased the ACL tear risk.^{140,213} The cross-sectional area of the ACL has been shown to be significantly smaller in women,^{165,169,214} and is related more to the height of the patient than the weight.¹⁶⁹ In addition, female ligaments show different mechanical properties, including lower stress and strain at failure, and a lower modulus of elasticity, even when size is accounted for.¹⁶⁸ This difference in ligament biomechanical properties can be explained by hormonal factors and differences in collagen genotypes. The AA genotype of the COL12A1 was shown to be associated with ACL injury,²¹⁵ while the CC genotype of COL5A1, and the TT genotype of COL1A1 have been shown to be protective.^{216,217} Increased Q angle is proposed to be a risk factor because a lateral tug on the quadriceps increases medial knee stress.²¹⁸ Hip varus and knee valgus can lead to increased loading of the knee, especially when landing from a jump.¹⁸³ Femoral notch measurements have been associated with ACL rupture; however, there is no sex difference in notch size.^{164,165,167,170} This injury risk seems to be related more to the size of the ligament, as notch volume has been shown to correlate with ligament size.¹⁶⁹ A larger femoral condyle offset ratio in combination with a larger lateral tibial slope is thought to increase anterior tibial translation and internal rotation, both of which increase strain.^{150,214,219,220} Despite these reports, no single specific anatomic risk factor has conclusively been noted to be a reliable predictor of noncontact ACL injury.

Hormonal influences as risk factors for noncontact ACL injury are a topic of continual debate. This group represents one of the more obvious sex differences, due to elevated serum levels of hormones such as estrogen, progesterone, and relaxin in females. Physiologic levels of the hormones estradiol and relaxin have been shown to decrease ligament strength and enhance soft tissue relaxation. Although ligament tissue contains hormone receptors, suggesting a potential role of hormonal fluctuations on tear rates,^{153,221} estrogen does not appear to have any negative effects on ligamentous mechanical properties.¹⁷¹ Relaxin, on the other hand, has been shown to increase protein expression responsible for collagen degradation, and decreased collagen synthesis, thus potentially lowering the ultimate stress to failure of the ligament.¹⁷² In addition, several proteins, such as agracan and fibromodulin, have been found to be upregulated in torn ligament tissue at the time of reconstruction.²²² It has also been suggested that hormonal influences on injury may be a result of neuromuscular changes during the menstrual cycle rather

TABLE 27.18 Summary of Risk Factors for Anterior Cruciate Ligament Injury

Anatomic

- Quadriceps angle,¹⁵⁵ knee valgus,^{136,156-158} and increased foot pronation,¹⁵⁹⁻¹⁶² have been associated with increased risk of anterior cruciate ligament (ACL) tear in some studies.
- The role of body mass index is not clear,^{163,164} but higher indices seems to be associated with injury risk.¹³²
- Femoral notch indices,¹⁶⁵⁻¹⁶⁷ ligament geometry,^{140,164,165,167,168} and tibial measurements are not definitively associated with increased risk of ACL tear; however, the smaller femoral notch in females, even when normalized to body weight, seems to be more associated with risk of injury.^{140,169,170}

Hormonal

- Sex hormones affect knee ligament laxity, but the overall effect is not clear.^{171,172}
- Injury rates have been noted to vary with the phase of the menstrual cycle.^{162,173-177}
- No evidence exists to recommend activity modification or restriction with respect to the phase of menstrual cycle to prevent ACL injury.

Environmental

- Harder ground, resulting from high evaporation and low rainfall, may increase the shoe traction interface.¹⁷⁸
- Increasing the shoe-surface coefficient may increase risk of injury (direct and alter movement (indirect)).¹⁷⁹⁻¹⁸¹

Biomechanical

- The ACL has different mechanical properties in women.¹⁶⁸
- Neuromuscular factors appear to be the most critical difference between adult men and women and are modifiable.¹⁸²
- The knee is part of a kinetic chain and as such is influenced by motion of the trunk, core, hips, and ankle.¹⁸³⁻¹⁸⁹
- Awkward landing, inability to recover from perturbed gait, and difficulty changing directions have been associated with increased risk of ACL injury.¹⁹⁰⁻¹⁹²
- Women exhibit less knee and hip flexion, increased knee valgus, increased internal rotation of the hip, and increased external rotation of the tibia when landing from a jump or changing direction.¹⁹⁰⁻¹⁹⁹
- Quadriceps-dominant activation patterns during deceleration, landing, and cutting have been noted in females.^{190,194,195,199,200-203}
- Decreased quadriceps stiffness and strength and decreased knee stiffness in females are thought to be associated with increased risk.^{176,204-206}
- Fatigue is associated with loss of dynamic control of the lower extremity.²⁰⁷⁻²¹¹
- Imbalance in strength, flexibility, and coordination is thought to be associated with increased risk of ACL injury.^{134,158,209,212}

than a direct effect on knee ligaments.^{156,221,223} Using urine and salivary hormone levels, as well as patient reported timelines, studies have shown that ACL tears are more likely to occur during preovulatory stages.^{151,173-175} However, during this time period the ACL demonstrates less laxity and increased stiffness, contradicting the theory that estrogen increases ligament laxity and enhances the risk for rupture.¹⁵¹ Hormonal intervention or activity modification based on the phase of the menstrual cycle is not recommended.

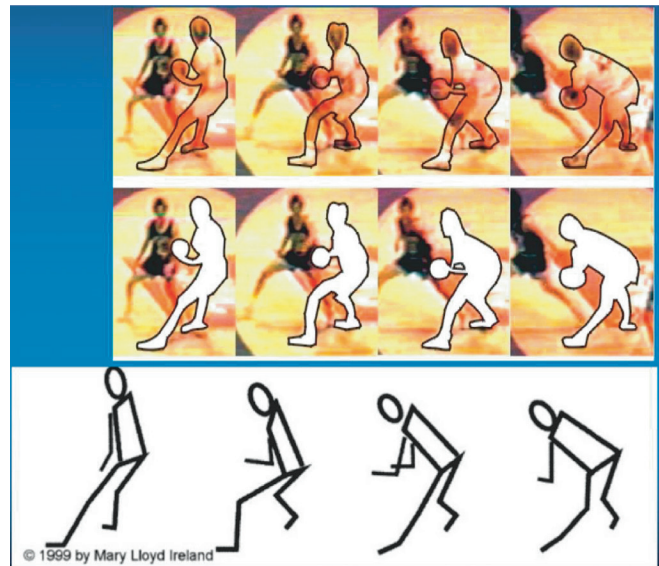


Fig. 27.4 Mechanism of noncontact anterior cruciate ligament injuries: female basketball player at the time of anterior cruciate ligament injury.

Modifiable environmental risk factors include shoes, surfaces, and shoe-surface dynamics. There has been much research regarding the interaction of the playing surface with the athlete's footwear.^{178-181,224-226} Playing surface material,^{179,181,226,227} as well as the environmental condition,¹⁷⁸ can influence the tear rate. Specifically, surfaces and conditions that provide higher traction have been shown to increase the risk of injury.^{179,180,225} However, researchers have yet to find the ideal shoe-surface combination that provides adequate traction for performance without increasing the risk of knee injury.²²⁴ Furthermore, the variation of athletic movement required in different sports combined with individual differences in style of play precludes the development of a single "perfect" shoe-surface combination that can be generalized to a wide range of athletic events.^{225,228}

Research on neuromuscular risk factors has revealed some of the most promising data on factors associated with ACL injuries; for example, 75% to 85% of noncontact injuries occur when landing from a jump, abruptly stopping, or pivoting to change direction with the knee in near full extension (Fig. 27.4).^{136,138,143-148,221,229} Muscle strength, coordination, recruitment patterns, endurance, and knee stiffness are all important variables for maintaining knee joint stability. Compared with men and normalized to body weight, women have significantly weaker quadriceps and hamstrings and have a diminished ability to produce adequate muscle stiffness across the knee.^{176,190,207} Additionally, elite female athletes tend to have quadriceps-dominant knees and forceful quadriceps contraction without appropriate hamstring co-contraction in response to the anterior tibial translation that increases ligament strain.^{182,221} Female athletes who sustain ACL injuries also have significantly greater peak valgus angles and moments around the knee.²³⁰ This observation has been validated by "drop jump" studies, which showed that in contrast to males, females achieved maximal hip adduction, knee valgus, and ankle eversion earlier in the stance phase and had less hip external rotation during landing; in addition, the

POSITION OF SAFETY			POSITION OF NO RETURN		
	Muscle activity	Body alignment		Body alignment	Muscle activity
Back		Normal lordosis		Forward flexed, rotated opposite side	
Hips		Flexed Neutral abduction/adduction Neutral rotation		Adduction Internal rotation	Flexors Adductors Iliopsoas
Knee	Extensors Abductors Gluteals				
	Flexors Hamstrings	Flexed		Less flexed, valgus	Extensors Quadriceps
Tibial rotation	Plantar Flexors	Neutral		External	Dorsi flexors
Landing pattern	Gastrocnemius	Both feet control		One foot out of control	Peroneals
	Posterior tibialis	Balanced		Unbalanced	Tibialis anterior

Fig. 27.5 Comparison of the position of safety with the position of no return that places the athlete at risk for anterior cruciate ligament injury.

angular velocity of knee valgus was nearly double.²³⁰ The center of gravity, postural adjustment, and motion patterns of the lower extremity kinetic chain affect loading at the knee.^{183-189,231} ACL load determinants include ground reaction force, body weight, muscle loading, knee stiffness, and the material properties of the ligament such as size, shape, and modulus.²³¹ Poor postural adjustment,^{189,192,193} combined with knee extension,^{190,191,193-195,232} valgus,^{190,191,193-195,232} and internal rotation,¹⁹¹⁻¹⁹³ increases the load placed on the knee (Fig. 27.5). To make matters worse, neuromuscular control is reduced in fatigued states,²⁰⁷⁻²¹¹ with less time to prepare for a cutting or landing movement.^{189,200,233,234} Numerous tests have been developed to identify those athletes with poor neuromuscular control.^{186,235-238} One such test, the single leg step-down test asks subjects to perform a series of controlled mini-squats. Athletes with poor performance on the test show some of the same kinematics seen at the time of ligament rupture (Fig. 27.6).²³⁹ Although more research is necessary, it is apparent that the combination of dynamic neuromuscular control patterns and inherent mechanical properties of the ACL are the principal contributing factors for increased injury rates in females.

Prevention of Noncontact Anterior Cruciate Ligament Injury

Identification of modifiable neuromuscular risk factors for non-contact ACL injury has fostered the development of successful, specific, preventative training programs to decrease knee loading and to improve protective movements in the lower extremity

TABLE 27.19 Measured Benefits of Anterior Cruciate Ligament Prevention Programs

- Decreased landing forces
- Decreased dynamic knee valgus
- Decreased varus and valgus movements of the lower extremity
- Increased muscular activation
- Improved movement patterns

kinetic chain (Table 27.19). Several studies have shown that these neuromuscular training regimens can reduce the risk of tear in specific patient groups.^{134,153,154,240-244} Many of these programs incorporate sport-specific drills to help athletes safely respond to unanticipated movement while emphasizing stretching, strengthening, aerobic exercise, agility drills, plyometrics, and risk awareness.²²⁴ Recent adjuncts to prevention programs include core, hamstring, and quadriceps strengthening exercises to decrease fatigue, and the education of athletes on appropriate landing and cutting techniques.¹⁸³ By training female athletes to alter their techniques and mechanics, these programs have demonstrated increased muscle activation and decreased landing forces, dynamic knee valgus, and varus and valgus moments.^{153,244} Although the long-term effectiveness of these programs is continually being evaluated, the early data and outcomes have shown immense promise, and some form of neuromuscular training is recommended for all at-risk individuals.

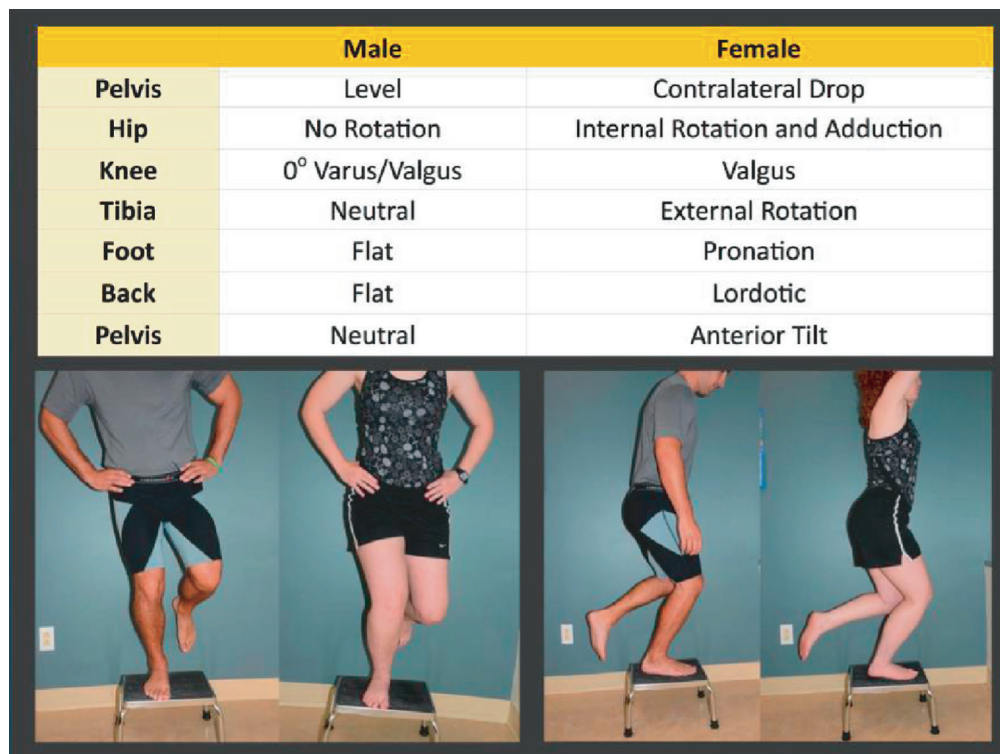


Fig. 27.6 Comparison of male and female strategies while performing a single leg squat.

Anterior Cruciate Ligament Reconstruction

Treatment guidelines for female athletes that sustain ACL tears are the same as those of their male counterparts, as overall results are similar between sexes.²⁴⁵⁻²⁵⁰ Regardless of the type of graft used, studies have demonstrated that women report improved knee function and less instability after reconstruction.^{245-249,251-259}

Although previous studies have cautioned the use of hamstring grafts in women,^{248,251,252,256} more recent studies have shown similar outcomes between graft type,^{249,259,260} with lower surgical morbidity and improved pain with hamstring grafts.^{255,258,259} The degree of postoperative anterior knee pain is variable, but less pain appears to occur after hamstring reconstruction in both men and women.²⁵⁵ Graft fixation, surgical technique, and post-operative rehabilitation are not sex specific.

After surgery, female athletes may struggle to return to their active lifestyles. In a retrospective analysis of soccer players, Brophy et al. found that after reconstruction, females were less likely to RTP, and more likely to have future ACL surgery than their male counterparts.²⁶¹ Unfortunately, ligament reconstruction also has not been shown to prevent posttraumatic osteoarthritis.^{262,263} Lohmander et al. evaluated female soccer players 12 years after an ACL injury.²⁶² Of the 84 women who answered questionnaires, 75% had knee-related symptoms that affected their quality of life and 51% had radiographic changes consistent with osteoarthritis. However, the development of posttraumatic osteoarthritis does not appear to be sex specific. After radiographic analysis of the MOON cohort, female sex was not identified as a risk factor for joint space loss at 2 or 6 years after surgery.^{264,265} In addition, in a large retrospective cohort study of patients with ACL tears, no sex differences were found in the long-term results

of reconstruction compared with nonoperative management. However, regardless of gender, surgery decreased the need for subsequent procedures and was associated with fewer meniscal and chondral lesions.²⁶⁶ Despite failing to prevent arthritis, ACL reconstruction provides stability and function, and decreases further intra-articular injury.

Frozen Shoulder/Adhesive Capsulitis

Idiopathic adhesive capsulitis (AC), also known as frozen shoulder, is a common condition that presents with pain and progressive loss of active and passive shoulder motion in the absence of a known intrinsic disorder.²⁶⁷ AC occurs more frequently in women between 40 and 60 years of age and is associated with diabetes, cardiovascular disease, thyroid disorder, Dupuytren contracture, as well as in those who have undergone breast cancer treatment.²⁶⁸⁻²⁷² AC is rare in females under 40 years of age unless they are diabetic. The nondominant shoulder is most commonly affected,²⁷³ with the contralateral shoulder affected in 20% to 30% of patients. There is an increased incidence of bilateral shoulder AC occurring in diabetics. This often occurs months to years after the original AC has resolved.²⁷⁴ AC is also less common in individuals who perform manual labor than those who have sedentary vocations.²⁷⁵

The prevalence of AC has been estimated to be between 2% and 5% with up to 70% of those patients being female.^{276,277} The rationale that explains gender disparity is ambiguous, and theories have been investigated, including hormonal fluctuations, autoimmune factors, genetic predisposition, and a causal associated with HSL-B27, have been inconclusive.²⁷⁸⁻²⁸⁰

Patients usually present with an insidious onset of vague pain at the deltoid insertion following minor or no trauma with

restricted external rotation and forward elevation. Older female athletes often present with several months of symptoms after the onset of pain and decreased motion begins to limit performance in their sport. On physical exam, there is not one specific nidus of pain; however, patients usually feel global shoulder pain. A decrease in the passive range of motion (ROM) in multiple planes in comparison to the contralateral shoulder is the hallmark of AC. This is initially best appreciated with external ROM with the elbow at its side and then with shoulder abduction with the scapula stabilized.

Radiographs are typically normal but important to obtain to rule out other pathology such as glenohumeral joint arthritis, calcific tendinopathy, or an unrecognized shoulder dislocation (particularly posterior dislocation). Magnetic resonance imaging (MRI) is often normal; however, it may show coracohumeral ligament thickening, rotator interval infiltration, or axillary recess thickening with edema.²⁸¹

The pathophysiology of AC is only partly comprehended. Histologic studies have shown inflammation of the subsynovial capsular layer. This inflammation leads to capsular thickening, fibrosis, and glenohumeral joint capsule tightening, with adhesion of the axillary pouch to itself and to the anatomic humeral neck.²⁸² Joint capsule inflammation also causes pain that is exacerbated by movement. Capsular biopsy reveals a chronic inflammatory infiltrate, an absence of synovial lining, and a moderate to extensive subsynovial fibrosis. Perivascular lymphocytic reactions also are seen.²⁸³

Synovial biomarkers also reveal that chronic inflammation is present in the shoulder joint. Kim et al. found elevated levels of intercellular adhesion molecule 1 (ICAM-1), a cytokine for leukocyte activation, proliferation, adhesion, and migration.²⁸⁴ Rodeo et al. found an increased presence of transforming growth factor- β (TGF- β) and TNF- α , two markers of chronic inflammation, in synovial fluid in patients with AC.²⁸⁵

There are four stages of AC. Stage 1 is the preadhesive stage where there is an inflammatory synovitic reaction without adhesive formation. During this stage the patient will have near full ROM of the shoulder; however, it is painful, especially at night. Stage 2 is the proliferation stage associated with early adhesion formation with increased loss of ROM. Stage 3 is characterized by decreased synovitis; however, more fibrosis occurs. It is in this stage that the axillary pouch is obliterated and adhered. Pain is usually less but the ROM is significantly decreased. In stage 4, the adhesions start to resolve and motion improves. Biopsies throughout these stages reveal a progression from mononuclear inflammatory infiltrates at stage 1 to reactive capsular fibrosis at stage 3 with resolution to normal capsule at the end of stage 4 (Table 27.20).^{270,286}

The natural history of AC is gradual symptomatic improvement. The time of progression through each stage is variable and the entire process can take 1 to 4 years for complete resolution.²⁸⁷ Conservative treatment has resulted in a patient satisfaction rate of 90% in one clinical study. In another study 50% of patients reported continued pain with 60% with motion deficits with conservative treatment.^{288,289}

Initial treatment of AC is nonoperative. No gender-related differences have been found in treatment or outcomes; however,

TABLE 27.20 Stages of Frozen Shoulder Syndrome

Stage	Phase	Timing (Months)	Presentation	Treatment
1		0–3	Pain, decreased motion	NSAIDs, PT $\pm$ injection
2	Freezing	3–9	Pain (more chronic), increasing loss of motion	NSAIDs, PT $\pm$ injection
3	Frozen	9–15	Pain lessening, significant motion loss	NSAIDs, PT $\pm$ injection, Possible surgery
4	Thawing	15–24	Resolving	Resolving

NSAIDs, Nonsteroidal antiinflammatory drugs.

compared with men, women initially tend to present with poorer external and internal rotation.²⁹⁰ The initial goal of treatment is analgesia and promotion of ROM. Gentle passive ROM with physical therapy or a home exercise program is the mainstay of treatment for AC. Formal physical therapy (PT) has not been shown to be superior to a home exercise program.²⁸⁷ Nonsteroidal antiinflammatory drugs have not been shown to improve pain or function compared to placebo for this condition. However, corticosteroids have demonstrated short-term analgesic efficacy with enhanced ROM.²⁹¹

Intra-articular corticosteroid injections have been found effective in providing short-term pain analgesia and improving the patient's tolerance of a stretching program.²⁹² It has been shown that a single intra-articular corticosteroid injection combined with a home exercise stretching protocol was effective in improving pain and ROM in patients with AC.²⁹³ In a prospective randomized trial, a corticosteroid injection applied before the beginning of a physical therapy program accelerated pain relief and recovery of function faster than an oral steroid alone.²⁹⁴ Three prospective randomized trials revealed no difference in pain relief or ROM between patients who received a glenohumeral or subacromial injection.^{295–297} While studies of corticosteroid injection have found significant short-term pain reduction and restoration of mobility, they have failed to find benefit of steroid injections past 24 weeks compared to control groups.^{296,298,299} A recent clinical study found that an ultrasound-guided extra-articular collagenase clostridium histolyticum injection with home exercise program has greater efficacy than a home exercise program alone in the treatment of AC. However, the use of collagenase may risk injury to other collagen-containing structures surrounding the shoulder.³⁰⁰

However, operative intervention should be reserved for patients who do not improve with PT or a home exercise program and in whom symptoms have been present for 6 months to 1 year. There is no consensus regarding the timing of a surgical procedure for AC other than patients should not be in the inflammatory phase of their disease. There is no evidence that early surgical intervention can shorten the recovery time of this disease.³⁰¹ Surgical treatments include capsular brisment, manipulation under anesthesia, arthroscopic capsular release, or open capsular release. Brisment is performed by injection 60 cc of saline into

the joint to create distension and rupture of the capsule.³⁰² Manipulation under anesthesia has an increased risk of iatrogenic fracture and has a greater efficacy in individuals who do not have diabetes.^{303,304} Arthroscopic capsular release is more controlled; however, meticulous dissection must be performed as the axillary nerve traverses close to the inferior aspect of the glenoid near the contracted axillary pouch. Manipulation is often performed after an adequate arthroscopic lysis of adhesions has been performed.^{305,306} A recent systematic review failed to show significant differences in outcome after manipulation under anesthesia versus arthroscopic capsular release.³⁰⁷ Open capsular release is rarely performed since the advent of arthroscopic techniques, and is usually only performed in patients with post-traumatic frozen shoulder who have retained hardware.

Shoulder Instability

The shoulder joint has a high functional ROM with a lack of inherent bony constraint. Glenohumeral joint stability is thus achieved through the interplay between the dynamic stabilizers (rotator cuff and deltoid muscles), static stabilizers (the labrum, glenohumeral ligaments, and joint capsule), and scapular kinesis. The shoulder has an inherent laxity, which is variable between individuals and allows for its normal physiologic motion.³⁰⁸ Female athletes with increased shoulder laxity are not predisposed to develop symptomatic shoulder instability.^{309,310}

“Laxity” is defined as increased normal physiologic motion of the glenohumeral joint, but it is not a pathologic entity.³¹¹ Shoulder instability is an abnormal or painful subluxation or dislocation of the glenohumeral joint. Shoulder instability has been described as voluntary, traumatic, or multidirectional instability (MDI). It has also been described in terms of the direction of instability such as anterior, posterior, or MDI. Neer described MDI as symptomatic laxity of the shoulder in two or more directions—one of which is inferior.³¹² MDI encompasses less than 10% of all patients with shoulder instability.^{313,314} Half of the patients diagnosed with MDI are female with no history of trauma.³¹⁵⁻³¹⁷ Half of those females with MDI also have generalized ligamentous laxity based on the Beighton scale.^{318,319} Increased laxity has not been directly correlated with concomitant multidirectional shoulder instability.³²⁰

MDI has been described in overhead athletes such as baseball players, tennis players, swimmers, and gymnasts.^{313,317,321,322} Atraumatic MDI is more common in the female athlete. A meta-analysis of shoulder instability in athletes revealed that 22% of unidirectional anterior instability, 27% of unidirectional posterior instability, and 35% of multidirectional instability were females.³²³

It has been proposed, but is debatable, that hormonal influences and decreased upper extremity muscle mass may predispose female athletes to MDI.^{308,324-326} It is now believed that MDI often occurs through micro-trauma to the shoulder capsule–ligamentous complex causing plastic deformation to the capsule and subsequent instability.^{308,312,327,328} Patients with MDI have significant imbalance of shoulder muscular control as well as alterations in shoulder kinematics and muscle activation during active ROM.^{329,330} Loss of shoulder proprioception as well as the loss of reflexive muscular protection against excessive humeral head translation is also seen in female athletes with MDI.^{328,331-333}

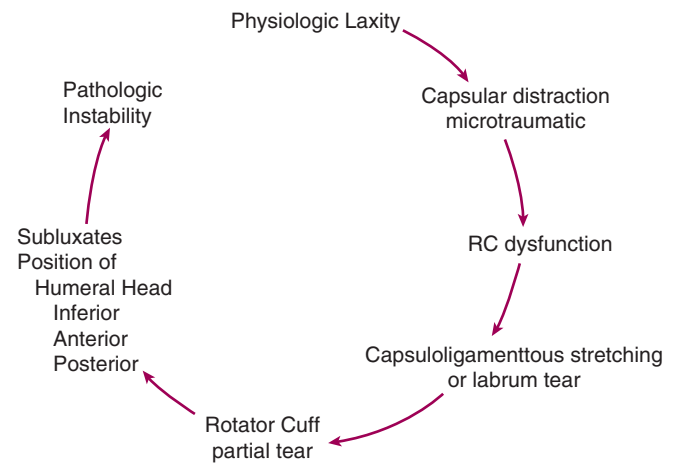


Fig. 27.7 Cycle of multidirectional instability in a shoulder.

Patients with MDI often have scapulothoracic dysfunction. Ideally, the scapula should move conjointly to keep the glenoid cornered underneath the humeral head. However, the scapula, in patients with MDI, droops laterally, increasing the inferior translation of the humeral head and increasing humeral head internal rotation leading to further instability (Fig. 27.7).^{334,335}

MDI often occurs during the second to third decades of life. The athlete will describe decreased strength with deteriorating athletic performance. They may also complain of fatigue, aching, pain at night, looseness, slipping, and sensory symptoms.³³⁶ Implicated sports include swimming (butterfly or backstroke), gymnastics, and overhead throwing sports.^{322,323,337} Patients often learn compensatory shoulder movements to avoid inciting activities.³³⁸

On physical exam, an anterior as well as posterior load and shift test should be performed. The amount of translation should be evaluated on a scale of 1 to 3 and documented.³³⁹ The sulcus test also should be performed with and without external rotation to evaluate the rotator interval (see Fig. 27.5). In addition, an anterior and posterior apprehension and relocation test should be performed. These tests are only positive if the test reproduces instability symptoms.³²⁴

It is important to obtain radiographs to identify any bony abnormalities of the shoulder, such as Hill-Sachs and bony Bankart lesions, glenoid version, dysplasia, hypoplasia and bone loss. However, x-rays are typically normal in patients with MDI. MRI arthrogram often reveals a normal labrum, biceps, rotator cuff, glenoid, and humeral head; however, a patulous capsule. MRI performed in the abduction external rotation (ABER) position may show the crescent sign or triangle sign secondary to a patulous capsule.³⁴⁰

Physical therapy for neuromuscular control, periscapular stabilization with strengthening, and rotator cuff strengthening are the primary treatments for MDI. The static stabilizers remain unchanged but improved muscle tone and proprioception leads to improvement in symptoms.³⁴¹ Burkhead reported good or excellent results in 83% of MDI patients treated with muscle-strengthening exercises.³⁴¹ More recently Misamore reported on a cohort of young athletic patients with MDI in which 52% had poor results and 25% had excellent results with physical therapy.³⁴² Despite these findings, a physical therapy program followed by

maintenance therapy is felt to be essential to maximize long-term outcomes in patients with MDI. Physical therapy should focus on improving the dynamic positioning of the humerus in the glenoid, instituting a proprioceptive exercise program, and improving the efficiency of the dynamic glenohumeral stabilizers. Rotator cuff strengthening can also improve active compression of the humeral head in the glenoid. The result is less shear force across the glenohumeral joint with improved centering of the humeral head in the glenoid.³⁴³

Those who fail at least a 3- to 6-month course of formal physical therapy may be offered surgical correction. Neer reported a 97% success rate of patients undergoing an open anterior inferior capsular shift with only one of 36 patients experiencing a resubluxation event after surgery.³¹² Baker reported a 2-year follow-up of 93% of good to excellent outcomes after an arthroscopic anterior inferior capsular shift with 86% of patients returning to sport.³³⁸ Thermal capsulorrhaphy, which once was popular, is no longer performed due to the increased risk of failure, chondrolysis, and thermal nerve injury.^{344,345}

Advances have been made in the diagnosis and treatment of MDI. Rehabilitation of the dynamic shoulder stabilizers is the gold standard of initial nonsurgical management. If this fails then surgical options can be used. Satisfactory outcomes have been obtained with either an open or an arthroscopic capsular shift. Treatment of patients with MDI can yield favorable results when there is careful examination, diagnosis, rehabilitation, and—when needed—appropriate surgical treatment.

Acetabular Labral Injuries and Femoroacetabular Impingement

There are many documented hip and groin injuries, which commonly affect athletes. Although in the past these injuries were often overlooked, improvements in diagnostic modalities including hip arthroscopy have led to improved identification of intra-articular hip pathology. Conditions, such as acetabular labral tears, capsular injuries, and intra-articular cartilage damage as a result of femoroacetabular impingement (FAI) and hip instability, have been increasingly recognized in athletes.³⁴⁶⁻³⁵¹

Just as our understanding of the complexity of shoulder pathology was enhanced by the ability to view the shoulder dynamically during arthroscopy, so has the development of better hip arthroscopic techniques resulted in a ballooning of our knowledge of hip pathology especially FAI. FAI is caused by abnormal hip mechanics, which result in abnormal contact of the femoral head within the acetabulum. Pincer impingement results from an excessive prominence of the anterolateral rim of the acetabulum, which in hip flexion “impinges” the labrum of the hip against the neck of the femur. This prominence of the anterolateral rim of the acetabulum can result from overgrowth of the anterior edge of the acetabulum or retroversion of the acetabulum. Repetitive impingement of the labrum injures the labrum with secondary injury to the cartilage of the adjacent acetabulum. Some report that pincer impingement is equally as common in males and females presenting in middle-aged adults (4th and 5th decades).^{352,353} However, a recent (2017) article by Shibata reported that in elite athletes pincer impingement was more common in females than males.³⁵⁴

Cam impingement occurs when a nonspherical femoral head, such as that which results from a prominence at the anterior superior head and neck junction of the proximal femur, rotates within the anterior superior acetabulum resulting in an injury to the articular cartilage at this location with secondary failure of the labrum. Cam impingement is more common in adolescent males than females (3:1 predilection).³⁵² Combined impingement can occur and, in elite athletes, was reported to be more common in males who also demonstrated greater acetabular cartilage damage secondary to this type of impingement. Bilateral symptomatic FAI is more common in males.²⁵³

Females have more issues with hypermobility of their hips than males and female ballet dancers have a greater incidence of hip dysplasia than do male ballet dancers.³⁵⁵ This dysplasia results in wonderful turnout, a positive for the dancer, but it can lead to FAI and early arthritis.

A relatively high incidence of dysplasia (21% of 126 hips) was also reported by Kapron et al. in collegiate females participating in a variety of sports with an additional 46% of female collegiate track and field, soccer, and volleyball athletes exhibiting borderline dysplasia.³⁵⁶

FAI typically presents similarly in males and females with the insidious onset of groin pain, which may be preceded by minor trauma, although many will report no history of a precipitating event. The pain is usually intermittent and exacerbated by physical activity and exercise. Although nonspecific, mechanical symptoms, such as locking and catching, also may be reported. Treatment of FAI, if the symptoms are stable, is nonoperative in females as in males. One must identify the offending activities and modify these. Squats often advocated in the weight room for obtaining strength should be avoided in these athletes as they can exacerbate symptoms. Strengthening of the muscles of the core may be helpful. If symptoms do not resolve, arthroscopic treatment of FAI is recommended. A recent study by Shibata and associates on elite athletes demonstrated the rate of highly competitive athletes returning to their former level of competitive sport participation after hip arthroscopy was similar for female and male athletes (84.2% of female athletes and 83.3% of male athletes).³⁵⁴

Idiopathic Scoliosis

Idiopathic scoliosis refers to the occurrence of a spinal curvature in the absence of congenital or neurologic abnormalities and is the most prevalent type of scoliosis accounting for approximately 70% of cases. Idiopathic scoliosis is further classified based upon on the age of appearance. Infantile scoliosis has an onset before 3 years of age; juvenile scoliosis has an onset between 3 and 10 years of age, and adolescent idiopathic scoliosis (AIS) occurs between age 10 and skeletal maturity. The adolescent form of this disease is most common and accounts for the majority of cases of idiopathic scoliosis. Although spinal deformities are concerning to both male and female athletes, AIS is of particular concern for females due to the 10:1 female to male predominance and the higher likelihood of progression in the female.³⁵⁷

The prevalence of AIS is reported to be between 2% and 3% in the general population.³⁵⁸ Although the etiology is largely unknown, current theories suggest a complex interplay

of genetic,^{359,360} biochemical signaling,³⁶¹ and neuromuscular control pathways.³⁶² A possible link between circulating levels of estrogen,^{363,364} life-style factors, such as nutrition and exercise level,³⁶⁵ and BMI^{366,367} also have been reported but more evidence is needed.

AIS has been reported to be more prevalent in female-dominated sports such as gymnastics, ballet, and swimming. Many athletes now begin vigorous training in childhood. Wojtys et al., found a correlation between the amount of exposure to intense athletic training and spinal curvatures in the immature spine.³⁶⁸ The authors felt that increased spinal curvature was associated with cumulative hours expended in training, and that overexposure to biomechanical stresses may play a role in progression.³⁶⁸

Warren et al. reported the incidence of idiopathic scoliosis to be 24% in ballet dancers.³⁶⁹ This was correlated to small body habitus and delayed menarche suggesting that hypoestrogenism and prolonged intervals of amenorrhea may predispose ballet dancers to scoliosis.³⁶⁹ A 10-fold higher incidence of scoliosis in rhythmic gymnastics trainees has also been reported, which was related to asymmetric loading, delayed menarche, and ligamentous laxity.³⁷⁰ Burwell et al. reported on the association of delayed puberty and scoliosis in rhythmic gymnasts and ballet dancers; however, they felt that the differing curve progressions were related to the type of training, suggesting the notion of sports-associated scoliosis.³⁷¹ Similarly, Becker found a 6.9% incidence of idiopathic scoliosis and a 16% incidence of mild curves in a study involving junior Olympic swimmers. The authors noted that curves were toward the swimmers dominant breathing side suggesting muscle imbalance as a possible contributor.³⁷²

Similar in female and male athletes, the diagnosis of AIS should be made after a thorough history and physical examination. The Adam's forward bend test is a common test used for screening. During this test the patient bends forward to touch the toes enhancing the spinal curve and demonstrating rib cage asymmetry. Particular attention should be given to the neurologic exam. Secondary sex characteristics should be assessed and the skin should be examined for café au lait spots, which are suggestive of neurofibromatosis. If significant deformity is present during screening, a radiographic evaluation, including a long AP and lateral film of the torso, should be performed. Routine MRI is typically not recommended; however, the presence of neurologic abnormalities or back pain that does not respond to several weeks of conservative care may warrant MRI evaluation.³⁷³

Stress Fractures

Studies in the 1970s and 1980s—the majority of which were done on the military population—reported that women have a higher incidence of stress fractures than men.³⁷⁴⁻³⁷⁸ In contrast, several recent studies in the athletic populations have shown a

modest increased risk or no increased risk of stress fractures in women compared with men.³⁷⁹⁻³⁸³ However, in 2004, following their survey of 5900 Division I collegiate athletes, Hame et al. found that although the incidence of all fractures was similar for men and women (0.0438 and 0.0461, respectively), the incidence of stress fractures was double for women and stress fractures were most common in track and field athletes.³⁸⁴ More recently Changstrom et al. reported that in a high school population girls sustained more stress fractures (63.3%) than boys (36.7%).³⁸⁵

Most agree that female athletes with menstrual irregularities have a higher incidence of stress fractures than female athletes with regular menstrual cycles.³⁸⁶⁻³⁹⁰ The reason for the relationship between abnormal menses and the increased incidence of stress fractures in athletic women is not well understood, nor is the literature consistent on the relationship between low BMD in amenorrhoeic athletes and stress fractures. Although one study in athletes with similar training schedules reported that athletes with stress fractures were more likely to have lower femoral neck and spine bone density, others report no association between the incidence of stress fractures and low BMD.³⁹¹⁻³⁹³ Both scenarios are reasonable since, in addition to hormonal variation, other variables that might influence the difference in the incidence of stress fractures between male and female athletes include bone geometric differences, the higher level of adipose tissue compared with lean body mass in females, and static and dynamic biomechanical differences between the sexes.³⁹⁴

As previously discussed in this chapter, a relationship has been defined between amenorrhea (loss of normal menstrual periods for over 3 months), low BMD and low energy availability—the so-called female athlete triad. Those with the triad have been reported to have an increased risk of stress fractures.^{43,395,396}

Treatment of the female athlete with a stress fracture should include, in addition to treatment of her fracture, an investigation into potential predisposing factors for stress injury. Overachieving (training errors), lack of sleep, and biomechanical considerations, and—more recently—vitamin D deficiency, have been linked to stress fractures in males. However, in females one must not only analyze the athlete's training schedules, sleep habits, vitamin D levels, shoe wear, lower extremity alignment, and other biomechanical considerations, but also her eating habits and menstrual cycle irregularities.⁴³ Is the athlete getting enough calcium, iron, and sufficient calories in her diet to support the energy needs of her sport? Does she have regular menstrual periods? If her menstrual periods are irregular, as often they are, referral to a sports gynecologist or to a family physician or internist skilled in the evaluation of menstrual dysfunction in athletes is appropriate. In females with recurrent stress fractures or stress reactions, an evaluation of BMD is recommended even in the absence of reported menstrual abnormalities.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

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